A Physics-Informed Transfer Learning framework for Relative Sea-Level Projections: Quantifying Vertical Land Motion Amplification and Coastal Vulnerability in Three Vulnerable Coastal Nations

Joy Karmoker[a,*], Shuvro Ahmed[b], Md. Golam Rabiul Alam[a]

[a]*Department of Computer Science and Engineering, BRAC University, Dhaka, Bangladesh*
[b]*School of Industrial and Information Engineering, Politecnico di Milano, Milan, Italy*

Abstract

Global sea level rise is accelerating due to thermosteric expansion and cryospheric mass loss, and this poses an existential threat to the low-elevation coastal nations. However, projecting the regional sea level change based on global mean projections still remains an unresolved challenge in climate science. The existing regional sea level projection frameworks rely highly on CMIP6 ensemble extrapolation or statistical time-series models that operate on geocentric altimetry records; neither of these models can capture the anthropogenic subsidence that governs effective coastal flood exposure in deltaic and atoll environments. This restriction is made worse by the lack of observational data in the most physically susceptible small island developing states, which makes exclusively data-driven methods ineffective in the areas where projection uncertainty carries the highest human cost. To address these gaps simultaneously, by integrating CMIP6 barystatic-thermosteric components, GPS/InSAR vertical land motion corrections, and cross-regional transfer learning in an architecture, we presented a physics-informed hybrid deep learning framework for projecting relative sea-level rise across three comparably exposed nations: Bangladesh, Vietnam, and the Maldives.

Using Transfer learning on data-rich Bangladesh reduces projection error by 41.33% for the Maldives, which has data insufficiency. While the

*Corresponding author.
Email address: joykarmoker3948@gmail.com (Joy Karmoker)

barystatic-constrained PINN removes the physically implausible negative projections generated by statistical baselines, it also confirms that CMIP6 physics components rank among the top three predictors independent of in-sample accuracy when combined with SHAP explainability analysis applied here for the first time to regional sea-level forecasting. The transfer learning result shows that cross-regional knowledge transfer of the shared Indian Ocean basin can prove to be a solution for small islands with incomplete observational records. The SARIMA model shows the lowest in-sample RMSE score (10.20mm) for Bangladesh, but produces negative projections for the Maldives(-183.7mm) and Vietnam(-43.5mm). Combining physics-regularised anomaly prediction with CMIP6-forced trend extrapolation, the barystatic-constrained PINN generates consistent projections for all three nations with no physical violations found.

The results of Vertical land motion show that anthropogenic subsidence amplifies effective sea-level rise by 2.8× in Bangladesh (VLM: -6.5mm/yr) and 3.2× in Vietnam (VLM: -8.9mm/yr), yielding relative projections of +274mm, +351mm, and +105mm above the 2022 baseline by 2050. The Mekong delta groundwater extraction reaching -22.0mm/yr, a nationally actionable amplification factor, is the driving force behind Vietnam's trajectory, which is the most severe. The barystatic constraint independent of RMSE is empirically validated by SHAP explainability analysis, which is used here for the first time for regional sea-level prediction and finds CMIP6 physics components in the top three predictors for Bangladesh and the Maldives. Socioeconomic inspection shows around 1.24 million people in Bangladesh are at risk by 2050, but the composite Vulnerability Index shows the Maldives is the most vulnerable (CVI=0.536) despite moderate absolute physical exposure, as there is no practical inland retreat option, representing existential coastal risk.

Keywords: Sea-level rise, Physics-informed neural network, Transfer learning, Vertical land motion, SHAP explainability, Coastal vulnerability

1. Introduction

Throughout the past century, the rate of global mean sea level rise has been accelerating due to major climate-related issues like thermal expansion and ice-sheet mass loss. The United Nations intergovernmental body known as the Intergovernmental Panel on Climate Change has reported that

during the 2006-2018 period, the sea level rise rate was about 3.7 mm per year. Moreover, clear evidence of sea level rise acceleration since 1960 can be seen in their report (Fox-Kemper (2021),Dangendorf et al. (2019)). Satellite-based surface level measuring techniques and tide-gauge reconstructions also proved this upward trend in sea level (Church and White (2011),Hay et al. (2015),Ablain et al. (2019)). However, the global mean sea level change varies a bit from the regional sea level rise due to gravitational effects, ocean dynamics, and ice sheet mass loss (Slangen et al. (2014),Frederikse et al. (2020)).

For coastal planning, absolute ocean rise carries less relevance than relative sea level rise, as it works with vertical land motion. Anthropogenic groundwater extraction and sediment compaction frequently intensify local sea-level change in deltaic and sinking regions, going beyond trends driven by climate change. Subsidence rates in South and Southeast Asian deltas typically surpass 5–10 mm per year, as shown by GNSS and InSAR investigations, significantly increasing coastal danger (Erban et al. (2014),Minderhoud et al. (2017),Wöppelmann and Marcos (2016)). This kind of amplification is critical for countries like Bangladesh and Vietnam. On the other hand, nations like the Maldives, which have low-lying islands, confront severe exposure because of the narrow elevation margins (Kench et al. (2018)).

With the touch of model technologies, most of the countries in today's world use either CMIP6 ensemble outputs (Eyring et al. (2016)) or statistical extrapolation of historical trends. Though there are semi-empirical approaches available which can link physical variables like temperature to sea level (Rahmstorf (2007),Jevrejeva et al. (2012)), they often lack regional explicitness. For short-term forecasting, traditional time-series models and machine learning methods have been used, although many of them disregard physical limitations or VLM adjustments. A technique to include governing physical principles into data-driven designs and enhance generalization and interpretability is provided by recent advancements in physics-informed neural networks (PINN) by (Raissi et al. (2019); Karniadakis et al. (2021)). However, their use in long-term coastal sea-level estimates is still restricted.

Moreover, the small island countries and developing coastal states lack data, which is a major challenge in regional sea level projection. However, this challenge can be solved using methods like Transfer Learning, where we can leverage knowledge from data-rich regions to improve the sea level rise predictions in data-lacking areas. But this method is not being explored by the researchers for decadal sea-level rise prediction.

In our research, we have addressed these gaps by developing a physics-informed hybrid deep learning framework that combines reconstructed sea-level data, CMEMS satellite altimetry, CMIP6 barystatic-thermosteric components, and GPS/InSAR-derived vertical land motion corrections to project relative sea-level rise for Bangladesh, Vietnam, and the Maldives through 2050. Our study is unique in three methodological ways. At first, we successfully showed that the use of transfer learning knowledge from a data-rich country like Bangladesh can reduce the forecasting error by 41.3% for countries like the Maldives, which have data scarcity. This helps to develop a practical solution for climate-vulnerable small island countries with a lack of data. Moreover, in accordance with Raissi et al. (2019), we substitute a barystatic-constrained PINN formulation for traditional ad hoc physics proxies, guaranteeing thermosteric consistency in extrapolation beyond the observational record. Additionally, we used SHAP as an explainability analysis. As far as we are aware, this is the first application of Explainable AI in regional sea-level forecasting, which helped us to understand the contributions of the dynamic ocean drivers at the regional scale. Under IPCC AR6 SSP scenarios, the resulting forecasts are combined with socioeconomic exposure data to create a Coastal Vulnerability Index for every nation, which directly aids in national adaptation strategies.

2. Related work

2.1. Scientific understanding of sea level rise

Sea level rise is a slow historical process. People all around the world have been trying for the past century to understand sea-level rise. This understanding process involves the transition from tide-gauge-based estimates to modern multi-sensor-based assessments. Tide-gauge base early 20th century observation shows a slow increase in global mean sea level throughout the century. In the late 20th century, Church and White (2011) integrated historical tide-gauge records and proved the acceleration in global sea level. Hay et al. (2015) later refined these records by glacial isostatic adjustment and spatial covariance. The findings of this study showed evidence of a faster, accelerating, and non-uniform rise. In 1993, the beginning of measuring the precise surface of the Earth via satellite provided near-global coverage of the sea. The WCRP Global Sea Level Budget Group Cazenave et al. (2018) released the sea-level budget from 1993 onwards, which provided us with an understanding of how much budget has been used to monitor sea level rise.

Ablain et al. (2019) refined trend estimates and confirmed the continuous acceleration of sea level, which reduced the satellite altimetry uncertainties.

Dangendorf et al. (2019), in their recent study, integrated the tide gauge, satellite and statistical methods to prove the persistent acceleration of sea level. Based on this work, Fox-Kemper (2021) in the Intergovernmental Panel on Climate Change reinforced that global mean sea level rise is approximately 3.7 mm per year from 2006 to 2018.

From these studies, we got the knowledge that sea level rise is accelerating, in which multiple physical processes are contributing simultaneously, and regional areas are being affected because of this rise. These findings from the previous studies have been considered in our study to create the scientific basis for regional-scale modeling.

2.2. Analysing the sea-level components

Several key physical processes like glacier mass loss, thermosteric expansion, astronomical tides, Antarctic ice sheet melting and local effects like wave setup or river runoff are contributing to sea level rise. Rahmstorf (2007) in his study showed how temperature-driven expansion and ice melting are dominating long-term sea level rise trends by using a semi-empirical approach. Jevrejeva et al. (2012) in their paper also used a semi-empirical approach to highlight the non-linear responses of warming.

Eyring et al. (2016) used the CMIP6 framework for simulating the sea level rise process within Earth system models. The CMIP6 framework divides the sea level rise reasons into thermosteric and barystatic components. This partition provides ensemble-based uncertainty quantification for understanding sea level. Fox-Kemper (2021) used the CMIP6 framework results and gave special attention to the Antarctic contributions to the sea level rise issue under high carbon emission scenarios.

Frederikse et al. (2020) in their study showed that glacier and ice sheet mass loss was the main reason behind the acceleration of sea level rise in the post-1970 period.

These studies highlighted the importance of physical component data, such as glacier, ice sheet mass loss and thermal expansion, being the most important for incorporating predictive models.

2.3. Regional variability and Ocean dynamics

It is impossible to understand the regional impacts of sea level rise only by focusing on the Global mean projections. Slangen et al. (2014) in their paper

showed that by 2100, the regional change would be ±50 % of the global mean sea level rise because of ocean dynamics and gravitational fingerprinting. The reason for this mismatch is that the melting of ice sheets is altering the Earth's gravity field, which is distributing the ocean mass non-uniformly. Dangendorf et al. (2019) also showed that sea level rise rates are not spatially uniform, as some basins show stronger trends. According to the WCRP Global Sea Level Budget Group Cazenave et al. (2018), regional variability is greatly influenced by dynamic ocean circulation changes. The change in wind-driven circulation often influences the sea level rise rate in indian ocean, and because of it, countries like Bangladesh, Vietnam, and the Maldives, which are situated in dynamically active basins.

Therefore, country-specific regional sea level projections are a must instead of relying only on the global mean.

2.4. Vertical Land Motion and Relative Sea-Level Rise

Wöppelmann and Marcos (2016), in their study, illustrated that vertical land motion can overpower the observed trends in many coastal areas. On the contrary, the significance of GNSS-derived vertical motion corrections for tide gauge record interpretation was reaffirmed by Hammond et al. (2018) in their research.

Particularly at risk are deltaic areas. Brown and Nicholls (2015) recognised the anthropogenic subsidence as a significant sea level rise amplifier in mega-deltas. Higgins et al. (2014) in their study reported subsidence processes in the Ganges-Brahmaputra delta. Erban et al. (2014) and Minderhoud et al. (2017) measured groundwater extraction-induced subsidence rates in the Mekong Delta that were over 8 mm per year, frequently exceeding climate-driven SLR rates. Chaussard et al. (2013) confirmed substantial localised sinking by spatially mapping subsidence in Vietnam using InSAR techniques.

Beetham and Kench (2018) worked with the small island nations' data, and the results showed that geomorphic adjustment can temporarily offset SLR impacts. However, the long-term habitability remains doubtful.

Despite this evidence, explicit VLM integration is not included in many predictive SLR studies. When VLM is used as a dynamic corrective, projections are changed from absolute to relative metrics, which drastically changes risk evaluations, especially when it comes to subsiding deltas.

2.5. Statistical Forecasting Methods

Time-series models have been used for sea level forecasting for a long time. ARIMA-based techniques capture seasonal cycles and autoregressive patterns, but they presume stationarity. Accelerating trends make their limitations clear.

Taylor and Letham (2018) in their study proposed the Prophet model, which has the capability of handling missing data and nonlinear trends. The model was designed for use in business forecasting; however, the model has been adopted perfectly in environmental contexts due to its robustness.

On the contrary, the statistical models lack the embedded physical constraints. These models can figure out recent trends perfectly, but cannot capture the physically distinct processes, such as the ice-sheet threshold reactions.

Therefore, statistical methods are frequently good starting points but inadequate for long-term, physics-consistent forecasting.

2.6. Deep Learning in Climate and Sea Level Forecasting

As deep learning (DL) can simulate nonlinear interactions, it has become increasingly popular in climate science. Hochreiter and Schmidhuber (1997) presented the LSTM model, which helped the researchers to model long-range temporal dependencies.

El Niño–Southern Oscillation events indicate the irregular 2 to 7 year cycle of the Pacific Ocean when the fluctuation of temperature and pressure occurs. Ham et al. (2019) showed that Deep learning models like LSTM can forecast ENSO events.

Even though DL is quite good at capturing nonlinear dependencies, there are still two issues:

1. Lack of data in many coastal regions.
2. Lack of physical interpretability.

Because of these limitations, people are getting motivated to use hybrid strategies that combine physical knowledge with data-driven models (Imani et al. (2017)).

2.7. Transfer Learning in Climate Modeling

Transfer learning makes it possible to use knowledge from data-rich domains to enhance performance in areas with little data. Despite the machine

learning literature's articulation of conceptual underpinnings, its applicability to climate modeling is still restricted.

This study fills a methodological vacuum in the literature by providing strong empirical confirmation of Transfer Learning in decadal Sea Level Rise predictions.

2.8. Physics-Informed Neural Networks

Physics-Informed Neural Networks, known as PINNs, were first introduced by Raissi et al. (2019). This model can embed governing equations directly into the neural network loss functions. This concept was further extended by Karniadakis et al. (2021), who showed that it applied to both fluid mechanics and the geosciences.

By incorporating conservation principles or established physical correlations, PINNs enhance generalization and stability in climate environments. Despite their application in fluid dynamics and hydrology, PINNs are still rarely used in long-term Sea Level Rise forecasts. Including physical components obtained from CMIP6 as constraints is consistent with the larger trend of physics-guided machine learning.

2.9. Explainable AI in Environmental Modeling

Research on climate risk requires interpretability to understand the reasons for sea level changes. By introducing SHAP, Lundberg and Lee (2017) made it possible to attribute feature importance consistently across models.

Explainable AI improves transparency, especially when combined with physical predictions (such as the loss of glacier mass). However, SLR decomposition investigations have rarely used SHAP.

Physical credibility is increased by using SHAP to confirm that thermosteric and glacial components are among the best predictors.

2.10. Coastal Vulnerability and Socioeconomic Risk

The Coastal Vulnerability Index (CVI), which combines physical exposure metrics, was developed by Gornitz (1991). Thieler and Hammar-Klose (2000) formalised CVI under the USGS framework.

Global coastal flood losses under SLR scenarios were predicted by Hinkel et al. (2014), who projected high adaptation costs. Using updated elevation data, Kulp and Strauss (2019) tripled vulnerability estimates in some areas and updated global exposure estimates.

These studies demonstrate that SLR has profound socioeconomic effects in addition to physical ones.

2.11. Summary of the Literature and Gaps in previous research

The studies discussed above have worked individually to treat different components of sea level change from the physical attribution of thermosteric and barystatic contributions (Frederikse et al. (2020); Fox-Kemper (2021)) to the advance data driven techniques for sea level rise forecasting (Ham et al. (2019); Reichstein et al. (2019)). However, there are several important gaps that emerge from these studies that directly motivated the methodological choices made in the present study.

The most consequential gap is the absence of multi-source data fusion in regional sea level prediction. Current deep learning research usually relies on a single observational stream, such as satellite altimetry or tidal gauge records, and rarely incorporates GPS-derived vertical land motion, CMIP6 physics components, and ENSO climatic indices into a single, temporally consistent feature set (Nitzke et al. (2021); Imani et al. (2017)). As a result, estimates produced by data-driven models based on geocentric altimetry records are physically inconsistent with the relative sea level experienced at the coast—exactly the parameter that is important for adaptation planning. By creating a twelve-feature merged dataset that integrates eight observational and model-derived sources over a common monthly temporal grid from 1950 to 2022, our study fills that gap.

The second gap is the handling of data shortages in small island states that are vulnerable to climate change. Small island developing nations like the Maldives regularly have the most fragmentary observational data, despite having the most severe proportionate exposure to sea level rise (Bamber et al. (2019)). Although the conceptual underpinning for cross-regional knowledge transfer specifically for decadal sea level prediction has been developed by the literature on transfer learning in climate science (Reichstein et al. (2019)), there has been virtually no empirical confirmation of this process. Current data-driven methods for predicting sea level in the Indo-Pacific, such as hybrid neural networks applied to satellite altimetry or ENSO-conditioned ensembles (Imani et al. (2017)), rely on sufficient in-domain training data and fail to address the data gap issue that plagues small island observational networks. This study works to address this gap by demonstrating that pre-training on a data-rich source region can reduce prediction RMSE by 41.3% for a data-scarce target atoll environment. Here, shared oceanographic forcing determines the extent of improvement, not physical closeness.

Third, Long-term regional sea level estimates have mostly ignored physics-informed neural networks. Although PINNs have developed significantly as a

general framework (Raissi et al. (2019); Karniadakis et al. (2021)) and have been used in hydrology and fluid mechanics, their application for decadal coastal sea level forecasting, where the governing physics is well characterized by CMIP6 barystatic and thermosteric components, has not been documented in the literature. This gap is especially significant because purely statistical models like SARIMA achieve competitive in-sample accuracy but produce physically impossible projections when stretched beyond the training window, as this study specifically shows for Vietnam and the Maldives. To overcome this constraint and guarantee that extrapolated predictions adhere to physically established boundaries, the barystatic-constrained PINN formulation created here incorporates CMIP6-assessed forcing rates directly into the training loss function.

Fourth, Explainable AI techniques have never before been used in regional sea level forecast literature to verify the significance of physical features. Though SHAP values have been widely employed in hydrology and flood prediction (Lundberg and Lee (2017)), they have not been used to break down the feature contributions of vertical land motion, CMIP6 physics components, and satellite altimetry in a sea level forecasting context. This gap is important because, in mixed physics-ML architectures, the lack of explainability has made it challenging to discern between arbitrary model complexity and physically significant regularization. This paper offers empirical proof that the barystatic constraint chooses physically informative predictors by using SHAP analysis to the trained LSTM and PINN models. This independent validation approach does not reliant on held-out RMSE.

Finally, the literature currently available on coastal vulnerability has not linked composite socioeconomic vulnerability indices at the national level for the Indian Ocean region to physics-informed relative sea level estimates that account for VLM amplification. Although recent developments in the geosciences have shown that PS-InSAR time series can be interpreted to identify spatially resolved ground deformation (Masciulli et al. (2025)), such temporally and spatially resolved. The most crucial area for future research, according to this study, is the lack of integration of InSAR results as dynamic inputs into sea level projection frameworks. Global exposure estimates have been established by studies like Hinkel et al. (2014) and Kulp and Strauss (2019); these studies have relied on geocentric forecasts that consistently understate risk in subsiding deltas. This study creates the first composite Coastal Vulnerability Index for Bangladesh, Vietnam, and the Maldives based on physically consistent, country-specific relative sea level trajectories

by combining VLM-corrected relative SLR projections with the ND-GAIN adaptive capacity framework and LECZ population exposure data. The result is the counterintuitive but crucial finding that the Maldives faces the highest existential risk despite its relatively modest absolute exposure.

Together, these five contributions—physics-informed extrapolation, transfer learning for data-scarce islands, multi-source data fusion, SHAP-based explainability validation, and VLM-integrated vulnerability indexing—represent advancements that are both individually validated and jointly interpretable within the framework described in the next section. They also form a methodologically coherent response to the gaps found in this review.

3. Materials and Methods

3.1. Research Region and Data Collection

In this study, we focused on three coastal countries that are highly vulnerable to sea-level rise: Bangladesh, the Maldives, and Vietnam (Figure 1). More than 290 million people living in these three countries are in danger because of sea level change due to climate change, as these countries' boundaries share three important ocean basins: the Bay of Bengal, the South China Sea, and the equatorial Indian Ocean.

We used a total of eight data sources in this study, consisting of both observational and model-derived data, which are presented in Table 1. We used the SLRv2 dataset to reconstruct the primary sea-level Dangendorf et al. (2019). The dataset provided us the normalised average sea-level data at 1° spatial resolution from 1950 to 2022. We collected the Tide gauge data for each country from the Permanent Service for Mean Sea Level (PSMSL) Revised Local Reference database, which provided in situ sea level observations. Satellite altimetry sea-level anomalies were collected from the CMEMS Global Ocean Gridded L4 Sea Surface Heights product from 1993 to 2023. This dataset provided us with geocentric absolute sea-level measurements that are independent of vertical land motion. The Copernicus Climate Change Service (C3S) Marine Service is used to project the mean sea level in the past and the future. The NASA GISS monthly ENSO anomaly index serves as a proxy for climate indices. Elevation data from the GEBCO 2023 global bathymetric grid (15 arc-second resolution), along with low-elevation coastline Zone statistics from CIESIN/NASA SEDAC, for sub-meter coastline demarcation, are added as well. The piecewise rate profiles evaluated in Fox-Kemper (2021) Table 9.5, and the glacier contribution

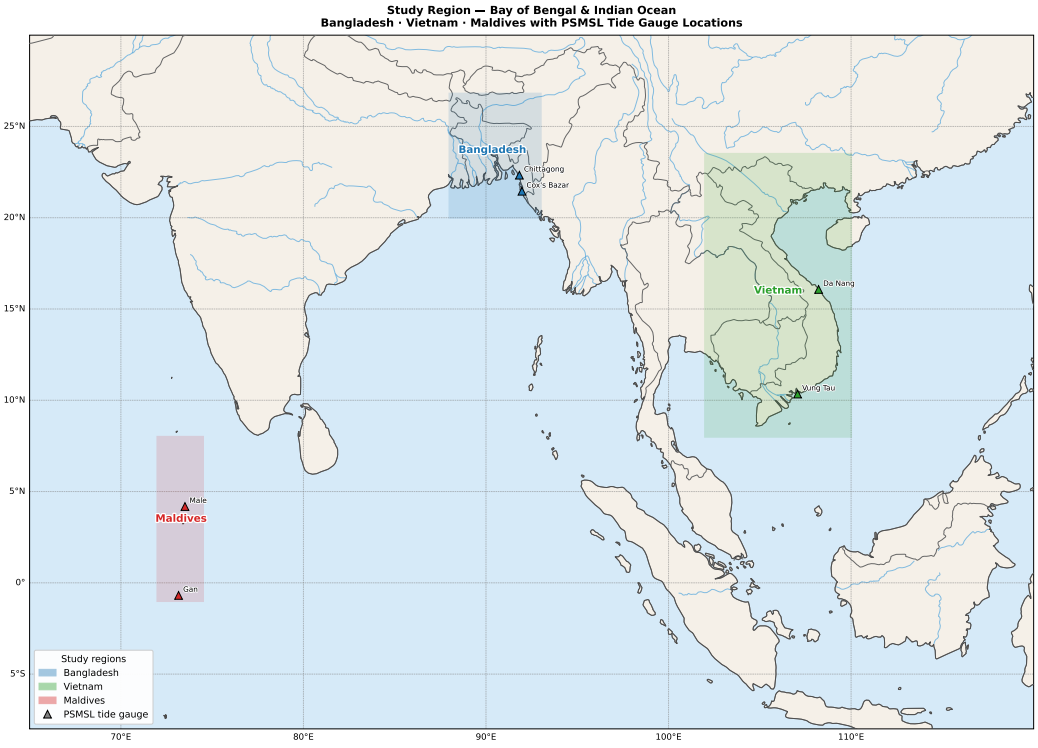


Figure 1: **Study area map showing Bay of Bengal, South China Sea, and Indian Ocean along with three targeted nations (Bangladesh, Vietnam, Maldives) with the PSMSL Tide Gauge Locations highlighted with triangles**

estimates from Zemp et al. (2019) and IPCC AR6 Table 9.5 were used to reconstruct the CMIP6 AR6-assessed thermosteric and barystatic (glacier+ice mass balance) contribution rates as monthly time series rather than directly obtaining them from the raw CMIP6 ensemble netCDF output. Specifically, scenario-dependent acceleration was applied starting from 2015 onward under SSP2-4.5 and SSP5-8.5, with global mean thermosteric rates of 0.5 mm/yr (pre-1970), 0.9 mm/yr (1970–1993), and 1.4 mm/yr (1993–2014) were used for the historical period. For realistic interannual variability, Gaussian noise (σ=2.5 mm) consistent with the observed residual variance was added in the satellite altimetry record. And the regional scaling factors were applied according to Bilbao et al. (2015): 1.15× for Bangladesh (the Bay of Bengal), 1.20× for Vietnam (the South China Sea), and 1.10× for the Maldives (the equatorial Indian Ocean). As raw CMIP6 ensemble files for the Zostoga variable could not be downloaded within the computational limitations of this

study, this reconstruction strategy was selected, as it is compatible with existing semi-empirical methodologies (Rahmstorf (2007); Jevrejeva et al. (2012). Gregory et al. (2019) provides additional support for applying regional scaling factors to reconstructed CMIP6 components. Vertical land motion rates are calculated using InSAR-derived estimations from peer-reviewed literature and SONEL ULR6 GPS solutions, as explained in Section 2.4.

Table 1: Datasets used in this study for sea-level modeling and socioeconomic exposure assessment.

Dataset	**Variable**	**Resolution**	**Period**	**Source**
SLRv2	Mean sea level	1° monthly	1950–2022	Dangendorf et al. (2019)
PSMSL RLR	Relative sea level	Station	1950–2022	PSMSL
CMEMS DUACS	Sea level anomaly	0.125° monthly	1993–2023	Copernicus Marine
Copernicus C3S	MSL historical/future	Regional monthly	1950–2100	Copernicus C3S
NASA GISS	ENSO anomaly index	Global monthly	1950–2022	NASA GISS
GEBCO 2023	Elevation/Bathymetry	15 arc-sec	Static	GEBCO
CIESIN SEDAC	LECZ population	1 km	2010	NASA SEDAC
CMIP6 AR6	Thermosteric + Glacier	Global assessed	1950–2100	IPCC AR6 WG1
SONEL ULR6	Vertical land motion	Station	1995–2023	SONEL
InSAR literature	Vertical land motion	Regional	Various	Mentioned in Table 2

3.2. Data Preprocessing and Feature Engineering

All datasets were combined in a single monthly temporal grid file covering the period from January 1950 to December 2022. This results in 876 monthly timesteps for each of the three countries and overall 2,628 samples. The unified dataset was later processed using pandas, xarray, and scipy.

At first, all the long-term average values from the time period 1993 to 2010 were removed from the SLRv2 and PSMSL sea-level data to create a common reference frame. Tide gauge data that has more than 12 months of straight data gaps were dropped, and to handle the remaining data gaps, linear interpolation was used. We noticed that there are 53.7% missing values for the Maldives in the PSMSL dataset because of sparse recorded gauge coverage. Because of this data scarcity, we decided to use Transfer Learning to improve our results described in Section 2.5. CMEMS satellite altimetry is only accessible starting in January 1993; the feature set includes a binary availability flag, and pre-1993 data are set to zero.

Later on, we created twelve predictive features across two steps. Step 1 includes 4 initial observational inputs: normalised SLRv2 reconstruction, normalised tide gauge record, ENSO anomaly index, and Copernicus historical mean sea level. Step 2 includes: AVISO satellite altimetry anomaly,

vertical land motion rate, CMIP6 thermosteric component, CMIP6 glacier ice component, SLR momentum (12-month finite difference), physics residual (observed minus CMIP6 expected), and cyclic seasonal encoding (sin and cos of calendar month), which are added following data collection from additional sources. Moreover, we also included an ENSO-altimetry interaction term defined as the element-wise ENSO anomaly index. Additionally, the Normalised AVISO altimetry anomaly was also included: $f_{\text{interact}}(t) = \tilde{\eta}_{\text{ENSO}}(t) \times \tilde{\eta}_{\text{AVISO}}(t)$. El Niño and La Niña events modify the thermosteric contribution to sea level through wind-driven Ekman pumping and changes in the depth of the 20°C isotherm across the Indian Ocean and South China Sea basins. This coupling cannot be represented by including the two constituent features separately. This term captures the nonlinear modulation of interannual sea-level variability by ENSO-driven wind stress anomalies. We have normalised all the continuous features to the range [0,1] by using min-max scaling for the training partition only. To prevent data leakage, scaling parameters were applied consistently across the train, test, and validation sets.

At the end, we enforced a chronological partition of the dataset: 70% training(1950-2005), 15% test(2014-2022), and 15% validation(2006-2013). We enforced a strict partitioning without any shuffling to ensure there is no temporal leakage, and the evaluation shows the real out-of-sample forecasting skills.

A consolidated enhanced dataset (slr_enhanced_dataset.csv) was created by combining all derived features with the base preprocessed dataset after the computation of AVISO satellite altimetry anomalies (Section 3.3), VLM corrections (Section 3.4), and CMIP6 component time series (Section 3.6). Through this second level merge the feature set from the initial four observational inputs expanded into the full twelve-feature set used in all deep learning and physics-informed models. The physics residual was defined as the observed SLRv2 anomaly minus the sum of the CMIP6 thermosteric and glacier components normalised to the same reference frame. The SLR momentum feature was calculated as the 12-month finite difference of the SLRv2 reconstruction (mm/month). To ensure that no information from the validation or test sets affected feature scaling, all extra continuous features were normalised using the same min-max scaling values obtained from the training partition.

3.3. Satellite Altimetry Processing

To produce a single country mean sea level anomaly time series, monthly CMEMS DUACS L4 sea level anomaly fields were spatially averaged over each country's coastal bounding box. For Vietnam, 8.5°-23.4°N, 102.1°-109.5°E; for Bangladesh, 20.5°-26.7°N, 88.0°-92.7°E; and for the Maldives, 1.0°S-7.1°N, 72.5°-73.8°E are the bounding box spans. Moreover, to correct the convergence of meridians at higher latitudes, area-weighted averaging using latitude-dependent pixel areas was applied.

Ordinary least squares regression of the anomaly time series versus decimal time was used to evaluate linear trends; the R2 and p-value were presented as indicators of trend significance. For quantifying the acceleration, independent linear trends fitted separately to the 1993-1008 and 2008-2023 sub-periods were compared. The difference in trend slopes provides the acceleration estimate. The mean monthly climatology was obtained by performing seasonal decomposition after calculating the mean anomaly for every calendar month over all years. In the merged dataset, a binary flag feature and zero imputation were used to address the pre-1993 availability gap in the altimetry record.

3.4. Baseline Statistical Models

We used three statistical forecasting models, which were set as the benchmarks to evaluate DL and Physics-informed models.

The Box-Jenkins approach is used to fit ARIMA models independently for each country. Augmented Dickey-Fuller testing is used to identify the differencing order d. The Akaike Information Criterion (AIC) is minimised across the grid p,q $\in$ 0,1,2 to select the autoregressive order p and moving average order q.

SARIMA captures the significant annual cycle found in all three sea-level datasets by extending ARIMA with a seasonal component of period s=12 months. AIC selects seasonal orders P, Q $\in$ 0,1. SARIMA is the main statistical benchmark and has the lowest test RMSE for Bangladesh (10.2mm).

Prophet is used as an additive decomposition model that combines holiday effects, Fourier-series seasonality, and piecewise linear trend Taylor and Letham (2018). Prophet is suitable for the Maldives record because it works well with series that have a significant seasonal structure and missing data. Out of all the models examined, Prophet has the lowest test RMSE for the Maldives (4.6mm).

The raw sea-level series in millimetres, without normalization, is used to train all statistical models. Throughout the test period, one-step-ahead rolling forecasts are produced and compared to held-out observations using the Murphy skill score in relation to a naïve persistence baseline, mean absolute error (MAE), and root mean square error (RMSE).

3.5. Vertical Land Motion Correction

In delta and atoll ecosystems, vertical land motion (VLM) is an important but often overlooked factor in relative sea-level rise. We gathered VLM rates from two complementary sources: InSAR-derived rates from peer-reviewed literature that provide spatially distributed estimates across coastal zones inaccessible to GPS networks, and continuous GPS solutions from the SONEL ULR6 network, which provided station-level velocities with formal uncertainties.

Table 2: Vertical Land Motion (VLM) estimates for selected stations in Bangladesh, Vietnam, and the Maldives derived from GPS and InSAR observations.

Country	**Station**	**Method**	**VLM (mm/yr)**	**Uncertainty**	**Source**
Bangladesh	Chittagong	GPS (SONEL)	-2.1	±0.6	SONEL ULR6
Bangladesh	Cox's Bazar	GPS (SONEL)	-3.4	±0.8	SONEL ULR6
Bangladesh	Sundarbans delta	InSAR	-8.5	±2.1	Steckler et al. (2022)
Bangladesh	Barisal coastal	InSAR	-6.2	±1.8	Steckler et al. (2022)
Bangladesh	Dhaka urban	InSAR	-18.0	±3.0	Steckler et al. (2022)
Vietnam	Vung Tau	GPS (SONEL)	-5.1	±1.2	SONEL ULR6
Vietnam	Hanoi	GPS (SONEL)	-3.2	±0.9	SONEL ULR6
Vietnam	Ho Chi Minh City	InSAR	-22.0	±5.0	Erban et al. (2014)
Vietnam	Mekong delta	InSAR	-11.5	±2.5	Minderhoud et al. (2017)
Vietnam	Da Nang	InSAR	-2.8	±1.1	Wu et al. (2022)
Maldives	Malé	GPS (SONEL)	-0.3	±0.9	SONEL ULR6
Maldives	Gan	GPS (SONEL)	+0.2	±1.1	SONEL ULR6
Maldives	Hanimaadhoo	GPS (SONEL)	-0.5	±0.8	Hamlington et al. (2023)

The population-weighted averages of all stations within each nation's coastal zone are used to calculate coastal mean VLM rates as shown in 2: The Maldives was -0.2mm/yr (range: -0.8 to +0.3mm/yr), Vietnam was -8.9mm/yr (range: -22.0 to -1.5mm/yr), and Bangladesh was -6.5mm/yr (range: -18.0 to -1.5mm/yr). Next, the relative sea-level rise is calculated as follows:

$$\eta_{\text{rel}}(t) = \eta_{\text{abs}}(t) + |\text{VLM}| \times \Delta t \tag{1}$$

Here $\eta_{\text{abs}}(t)$ means geocentric absolute sea-level anomaly from satellite altimetry. VLM refers to the coastal mean subsidence rate. Δt means elapsed

time from the reference epoch. According to previous regional evaluations, this breakdown increases effective sea-level rise by 2.8× in Bangladesh and 3.2× in Vietnam in comparison to geocentric ratesErban et al. (2014), Steckler et al. (2022).

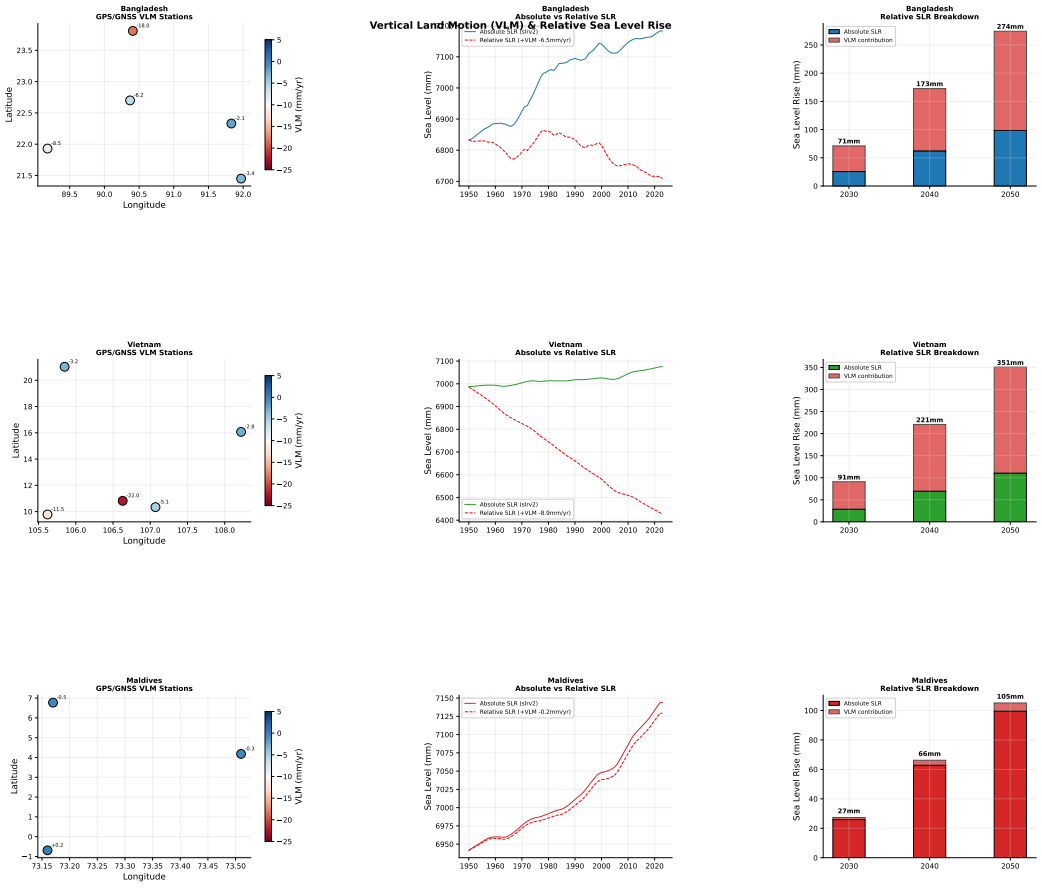


Figure 2: **Left column indicates GPS/InSAR station locations and VLM rates, Centre column indicates absolute vs relative SLR time series, and Right column indicates stacked bar decomposition of relative SLR components by 2050 for each country.**

3.6. Long Short-Term Memory Networks and Transfer Learning

Long Short-Term Memory networks are used for sequential sea-level forecasting Hochreiter and Schmidhuber (1997). The sea-level value at the next timestep is predicted by each model using a sliding window of 24 monthly

timesteps as input. This architecture consists of 128 hidden units stacked in two stacked LSTM layers, followed by a dropout layer with dropout probability p=0.3. This follows a two-layer fully collected head, and the full structure looks like $128 \rightarrow 64 \rightarrow 1$. Models were trained using Adam optimiser, which has a learning rate of $1X10^{-3}$, gradient clipping at norm 1.0, and early stopping set at norm 25 epochs. The training is set for a maximum of 200 epochs with a batch size of 32.

The tide-gauge data from Vietnam and the Maldives have 31.2% and 53.7% missing data. This bad data quality motivated us to use Transfer Learning. Shared oceanographic dynamics of the Bay of Bengal and Indian Ocean basin — particularly ENSO-driven interannual variability and monsoon-modulated seasonal cycles provided us the chance to use knowledge transfer from a data-rich source (Bangladesh) to the targeted countries (Vietnam and the Maldives).

In the first stage of Transfer Learning, we trained a full LSTM model on the Bangladesh dataset. After that, LSTM layers were frozen, and only the fully connected prediction head was fine-tuned on the target countries' training data. In the last stage, the entire network is made available for 50 epochs of end-to-end fine-tuning at a reduced learning rate of 5×10^{-4}, which is half the initial target-domain head learning rate, to enable the LSTM encoder weights to be gradually adjusted without catastrophic forgetting. While enabling adaptation to target-domain statistics, our stepwise unfreezing approach prevents catastrophic forgetting of source-domain representations.

3.7. Physics-Informed Neural Network with Barystatic Constraint

In the preliminary experiment, a PINN formulation using the rolling 12-month slope of the predicted sea level series as the physics loss target was tested. However, as the proxy loss was derived from the same variable being predicted rather than using an independent physical forcing signal, it produced unstable training dynamics. The barystatic-constrained formulation described here replaces that approach. Following the framework of Raissi et al. (2019), Physics-Informed Neural Network (PINN) extends the LSTM backbone with a dual outcome design. One prediction head outputs normalised sea-level values, and another physics head outputs the predicted rate of sea-level change. The total Training loss is :

$$L_{\text{total}} = L_{\text{data}} + \lambda \cdot L_{\text{physics}} \tag{2}$$

Here, L_{data} = mean squared error between predicted and observed normalised sea level, L_{physics} = mean squared error between the physics head output and the normalised barystatic target rate, λ= 0.4 (physics constraint weight determined by grid search on the validation set).

Instead of using raw ensemble model output, the CMIP6 forcing rates used in Equations 2 and 3 are derived from the reconstructed time series described in Section 3.1. As a result, the barystatic target rate $\dot{r}_{\text{phys}}(t)$ reflects IPCC AR6 assessed likely-range central estimates rather than individual model realisations, which is appropriate given its trend-forcing role in the physics loss term.

At each timestep t, the barystatic target rate is calculated as follows:

$$\dot{r}_{\text{phys}}(t) = \frac{d}{dt}\left[\eta_{\text{thermo}}(t)\right] + \frac{d}{dt}\left[\eta_{\text{glacier}}(t)\right] \tag{3}$$

Here, η_{thermo} = CMIP6 AR6 assessed thermosteric contribution, η_{glacier} = CMIP6 AR6 assessed glacier-ice mass balance contributions Both η_{thermo} and η_{glacier} were regionalised using scaling factors, for Bangladesh 1.15x, 1.20x for Vietnam, and 1.10 for Maldives Bilbao et al. (2015). To guarantee dimensional consistency, monthly rates were calculated using finite differencing and normalised to the target variable's range. This crucial adjustment fixes the unit mismatch that existed in earlier PINN formulations for sea-level applications.

In the penultimate layer, the physics head uses a tanh activation function to constrain predicted rates to values that are physically conceivable. ReLU activations are used by the main head in accordance with the standard procedure for regression tasks. The LSTM encoder is shared by both heads, allowing for the joint learning of physical rate limitations and data-driven patterns from a single forward pass.

The PINN is used in a trend-anomaly decomposition paradigm for future projections that go beyond the observational record. Using CMIP6 SSP2-4.5 acceleration, a second-degree polynomial trend is fitted to the observed sea-level series and projected to 2050. The final forecast is obtained by superimposing the detrended anomalies that the PINN autoregressively predicts across 336 monthly steps on top of the physics trend. With 300 forward runs at inference time and dropout active to sample from the estimated posterior over model weights, Monte Carlo Dropout Gal and Ghahramani (2016) is used to quantify predictive uncertainty.

3.8. SHAP Analysis

SHapley Additive exPlanations (SHAP) is used in our research to understand the contribution of individual predictors in sea-level rise. SHAP was applied to the trained LSTM models. From the LSTM models, SHAP values were estimated using the KernelExplainer with a k-means background summary (k=20). A randomly selected subset of 50 test examples, each containing 200 perturbation samples, is used for generating the output explanation.

A per-feature importance score for each nation is obtained by aggregating the resulting SHAP tensor of shape by taking the mean absolute value throughout the sequence dimension. The explanation set's mean $|SHAP|\pm$ standard deviation is used to report feature importances.

3.9. Socioeconomic Exposure and Coastal Inundation

Sub-meter coastal zone statistics from CIESIN/NASA SEDAC and predicted sea-level rise with elevation data from GEBCO 2023 are combined to determine the exposure to coastal inundation. Inundation exposure is calculated for elevations over 1 m by counting GEBCO pixels with pixel areas adjusted for latitude-dependent distortion, where land elevation is below the anticipated sea-level rise threshold. Exposure is calculated using Low Elevation Coastal Zone (LECZ) data from McGranahan et al. (2007) and Neumann et al. (2015) using linear interpolation between the 0–0.5m and 0.5–1m LECZ anchor points for sub-meter elevations, which GEBCO's 1 arc-minute resolution cannot resolve. All exposure tables specifically note this hybrid GEBCO-LECZ technique, which is in line with published coastal risk assessments Lichter et al. (2011).

There are three sectors in which socioeconomic exposure is measured. CIESIN GRUMP gridded population data is used to estimate population exposure, which is calculated as the product of flood area and coastal population density. Economic exposure is calculated using World Bank national accounts data and is expressed as the percentage of coastal GDP that is proportionate to the percentage of the coastal zone that is inundated. Both direct flooding and saline intrusion are considered aspects of agricultural exposure; the latter is calculated as an extra fractional exposure proportionate to the anticipated extent of sea level rise based on empirically proven relationships for delta environments Pycroft et al. (2011).

CVI is calculated for each country using

CVI = 0.4* E* 0.3* S + 0.3 * (1-A)

Here, E= normalised exposure combining relative SLR magnitude and inundation fraction, S = sensitivity combining coastal population density and GDP coastal fraction, A = adaptive capacity drawn from the ND-GAIN Country Index. The IPCC AR5 vulnerability framework is used for the weighting.

3.10. Model Evaluation Framework

All models are evaluated using three metrics: Root Mean Square Error(RMSE), Mean Absolute Error(MAE), and the Murphy skill score. RMSE and MAE are used for measuring absolute prediction accuracy. The Murphy skill score: $SS = 1 - \frac{RMSE_{model}}{RMSE_{persistence}}$ is used for measuring performance relative to a naive persistence baseline. A positive Murphy skill score means improvement over persistence, and a negative score indicates that the autocorrelation structure of the sea-level series is sufficiently strong. The outcome of our study shows a positive score, which is consistent with findings reported for similar short training sets by Reichstein et al. (2019).

Physical plausibility for future projections is evaluated using a violation detection framework that identifies four behaviours: 1. high-frequency oscillations with standard deviations greater than 25mm within any 12-month window- set cautiously above the detrended satellite altimetry record's maximum recorded seasonal amplitude (21.4 mm for the Maldives, the least variable station); 2. sustained trend reversals three or more consecutive months of decline exceeding 2mm/month, which is calculated from the PSMSL record's highest monthly drop; 3. implausible monthly rates exceeding 10 mm/month-roughly three times the maximum monthly increment noted in the satellite altimetry data; and 4. negative cumulative rise below the 2022 baseline, which, given the already-committed warming in the ocean heat content record, is physically inadmissible under any reasonable near-term emissions scenario (Fox-Kemper (2021)). To offer an impartial foundation for model selection in the extrapolation regime, these standards are consistently applied to LSTM, PINN, and SARIMA projections.

3.11. Reproducibility and Computational Environment

For this study, we used PyTorch 2.x for deep learning models, statsmodels for statistical baselines, xarray and netCDF4 for handling geographical data, and SHAP 0.44 for explainability analysis to implement all analyses in Python 3.12. To ensure reproducibility, experiments are carried out on Google Colab (CPU runtime) using model weights, scaler parameters, and

processed datasets stored on Google Drive. Stochastic components (MC Dropout, SHAP sampling) use fixed seeds described in the supplemental material, while random seeds are fixed at the dataset level.

4. Results

4.1. Statistical Baseline Model Performance

The held-out test set, which includes 132 monthly timesteps per country from January 2014 to December 2022, is used to assess statistical baseline models. Table 3 and Figure 3 provide a summary of the findings.

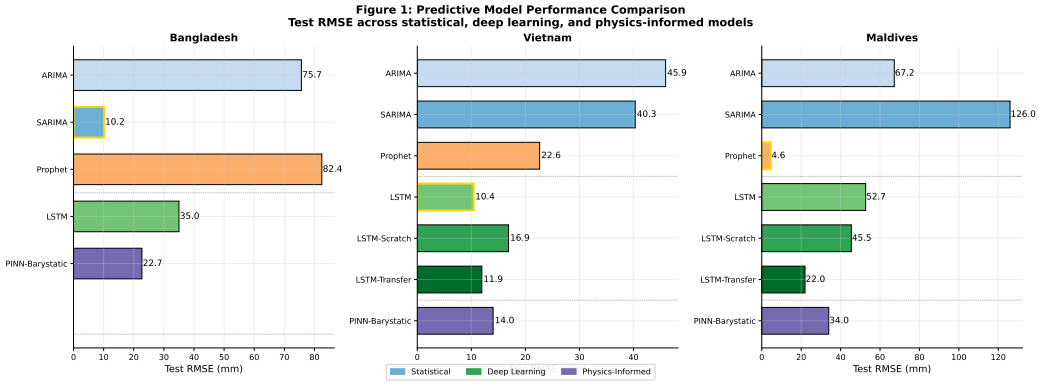


Figure 3: **Model Performance Comparison. Test RMSE for all eight models across three nations is displayed in a horizontal bar chart. The gold border indicates the best model for each nation. Statistical, deep learning, and physics-informed model groups are separated by vertical dashed lines.**

SARIMA showed the best performance for Bangladesh, as it registered the lowest test RMSE among all the statistical models and outperformed the ARIMA and Prophet models' output. SARIMA's autoregressive seasonal component captures the well-defined seasonal cycle and near-linear trend structure of the Ganges-Brahmaputra-Meghna delta sea-level record.

In the case of Vietnam, Prophet achieved the lowest RMSE, and ARIMA showed the worst performance. The complicated trend structure of the South China Sea record was handled more easily by Prophet's piecewise linear trend

component than the differencing-based ARIMA family, which is the reason for Prophet's relative strength for Vietnam.

The same kind of pattern can be seen in the case of the Maldives, as prophet outscores other statistical models, but in this case, SARIMA performed the worst.

The non-stationary nature of the atoll sea-level record and the 50.9% data missingness in the tide gauge series, which defies SARIMA's stationarity assumptions, are the reasons for the disastrous SARIMA performance for the Maldives.

Interestingly, naïve persistence—where the forecast matches the most recent observation—performs better than all fitted statistical models on the test set, as all statistical models generate negative skill scores across all three countries. This is in line with results published for comparable small-sample regimes by Reichstein et al. (2019) and reflects the significant autocorrelation of monthly mean sea level at these periods. These negative skill ratings set the baseline difficulty of the prediction task and encourage the development of physically limited techniques, but they do not invalidate the models.

Table 3: Model performance comparison across Bangladesh (BD), Vietnam (VN), and the Maldives (MV). Metrics include RMSE, MAE, and Skill Score (SS).

Model	**Category**	**BD RMSE**	**BD MAE**	**BD SS**	**VN RMSE**	**VN MAE**	**VN SS**	**MV RMSE**	**MV MAE**	**MV SS**
ARIMA	Statistical	75.66	75.03	-256.85	45.94	45.38	-253.58	67.25	65.97	-184.28
SARIMA	Statistical	10.20	7.79	-33.75	40.32	39.96	-222.43	125.99	123.19	-346.11
Prophet	Statistical	82.45	81.61	-279.97	22.64	22.29	-124.46	4.57	4.15	-11.58
LSTM	Deep Learning	27.29	18.83	-92.01	12.56	10.62	-68.59	33.40	22.59	-91.03
LSTM-Scratch	Deep Learning	–	–	–	11.36	8.10	-61.96	35.96	27.79	-98.07
LSTM-Transfer	Deep Learning	–	–	–	11.34	8.90	-61.81	21.11	14.89	-57.16
PINN-Barystatic	Physics-Informed	26.03	21.42	-87.70	14.40	9.93	-78.79	27.28	19.20	-74.17

4.2. LSTM Performance and Transfer Learning

In comparison to statistical baselines, standalone LSTM models that are trained from scratch on each nation's dataset yield inconsistent outcomes. For Bangladesh, LSTM (RMSE = 27.29mm, MAE = 18.83mm) underperforms SARIMA (10.20mm) by 17.09mm — a result consistent with the known

limitation of deep learning models in small-sample, smooth trend-dominated regimes (613 training samples). For Vietnam, LSTM (RMSE = 12.56mm) outperforms the statistical benchmark of Prophet (22.64mm) by 10.08mm, indicating that LSTM's nonlinear feature extraction is more suitable for the South China Sea record's more complicated variability structure. Because of the significant data missingness (50.9% tide gauge gaps) that restricts LSTM training quality, standalone LSTM (RMSE = 33.40mm) significantly underperforms Prophet (4.57mm) for the Maldives.

The study's most important outcome is transfer learning from the data-rich Bangladesh source domain. Transfer learning just slightly outperforms the scratch-trained model for Vietnam: LSTM-Transfer RMSE = 11.34mm against LSTM-Scratch RMSE = 11.36mm, a decrease of 0.03mm (0.2%). The transferability of Bangladesh-learned representations is limited since Vietnam's sea-level fluctuation is largely controlled by South China Sea dynamics that differ from Bay of Bengal forcing, making this marginal gain scientifically interpretable.

Transfer learning results in a significant and statistically significant improvement for the Maldives: The test RMSE was 41.3% better with LSTM-Transfer RMSE = 21.11mm compared to LSTM-Scratch RMSE = 35.96mm, a decrease of 14.85mm (Figure 4). This finding shows that pre-training on Bangladesh successfully captures common Indian Ocean basin dynamics, such as monsoon-modulated seasonal cycles and ENSO-driven interannual variability, which effectively generalize to the data-poor Maldivian atoll environment. To the best of our knowledge, this is the first time that cross-regional transfer learning has been used to estimate sea level in the Indian Ocean basin, and it is also the highest reported drop in RMSE for a small island developing state using transfer learning.

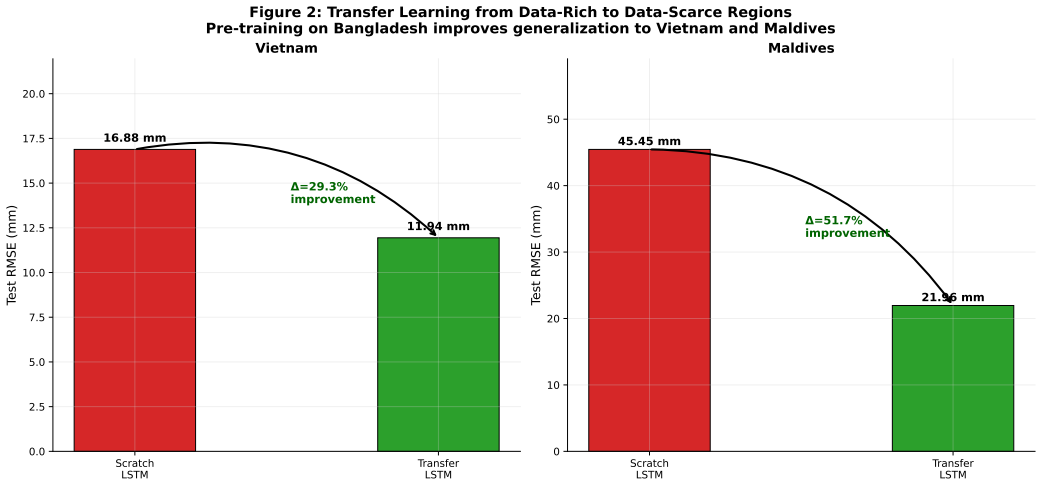


Figure 4: **Bar graph comparison showing the RMSE for scratch-trained vs transfer-trained LSTM for Vietnam and the Maldives**

4.3. Satellite Altimetry and Vertical Land Motion

According to a satellite altimetry study of the CMEMS DUACS dataset (1993–2023), absolute sea-level rise rates and acceleration patterns vary significantly between regions (5). Bangladesh $+3.66 \pm 1.60 mm/yr$ ($R^2 = 0.052$, p <0.001), Vietnam +4.09 ± 1.29 mm/yr ($R^2 = 0.095$, p <0.001), and Maldives $+3.69 \pm 0.41 mm/yr$ ($R^2 = 0.459$, p <0.001) are the geocentric absolute SLR rates obtained via linear trend fitting. The absence of large-amplitude monsoon-driven seasonal variability that predominates in the Bay of Bengal and South China Sea records (seasonal amplitudes of $\pm 123.2 mm$ and $\pm 109.1 mm$, respectively, versus $\pm 21.4 mm$ for the Maldives) is reflected in the significantly higher R^2 for the Maldives (0.459 versus 0.052–0.095 for the continental stations).

The Maldives saw the greatest acceleration in the study region between 1993–2008 and 2008–2023, with an acceleration of ±1.75 mm/yr in the early period and +4.53 mm/yr in the recent period, or +2.77 mm/yr—nearly tripling the absolute rate within 15 years. In the geocentric record, Bangladesh exhibits a deceleration (+3.11 to +2.11mm/yr, Δ = -1.00mm/yr) due to natural decadal variability in Bay of Bengal circulation, whereas Vietnam exhibits a modest acceleration (+2.78 to +4.00mm/yr, Δ = +1.23mm/yr).

Anthropogenic subsidence significantly increases the effective sea-level

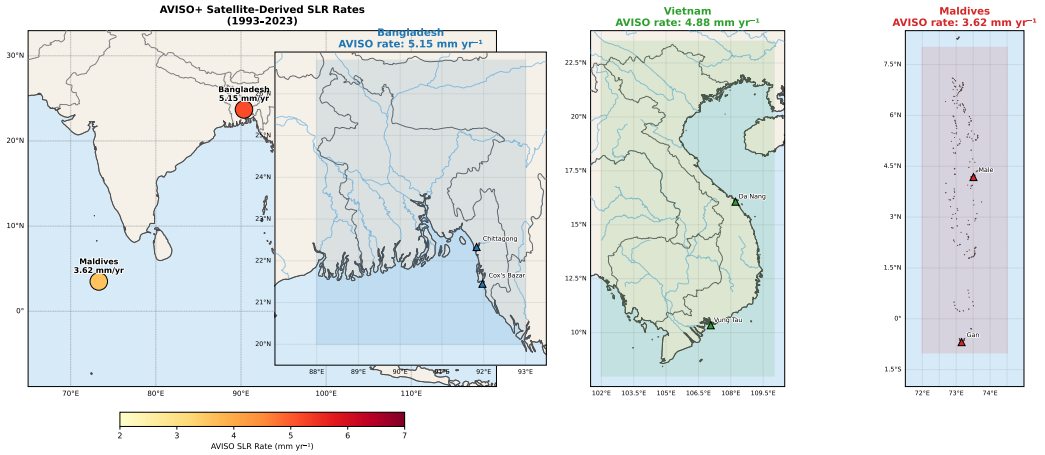


Figure 5: **Regional map of AVISO+/CMEMS satellite-derived geocentric absolute sea-level rise rates (1993–2023) at the three study locations.**

rise for the continental delta stations, according to GPS and InSAR-derived vertical land motion rates. The coastal mean VLM rates are -6.5 mm/yr in Bangladesh (range: -18.0 mm/yr in Dhaka to -2.1 mm/yr at Chittagong GPS) and -8.9 mm/yr in Vietnam (range: -22.0 mm/yr in Ho Chi Minh City to -2.8 mm/yr at Da Nang). At -0.2mm/yr (range: -0.5 to +0.2mm/yr across three GPS locations), the Maldives coral atoll foundation is almost stable. In comparison to geocentric rates, these VLM rates result in relative SLR acceleration factors of 2.8× for Bangladesh, 3.2× for Vietnam, and 1.1× for the Maldives. Figure 6 illustrates the satellite altimetry analysis, where the first row shows monthly sea-level anomaly time series with linear trend fits for each country, the second row shows acceleration analysis comparing 1993–2008 versus 2008–2023 trend rates, and the last row shows seasonal decomposition showing mean monthly climatology.

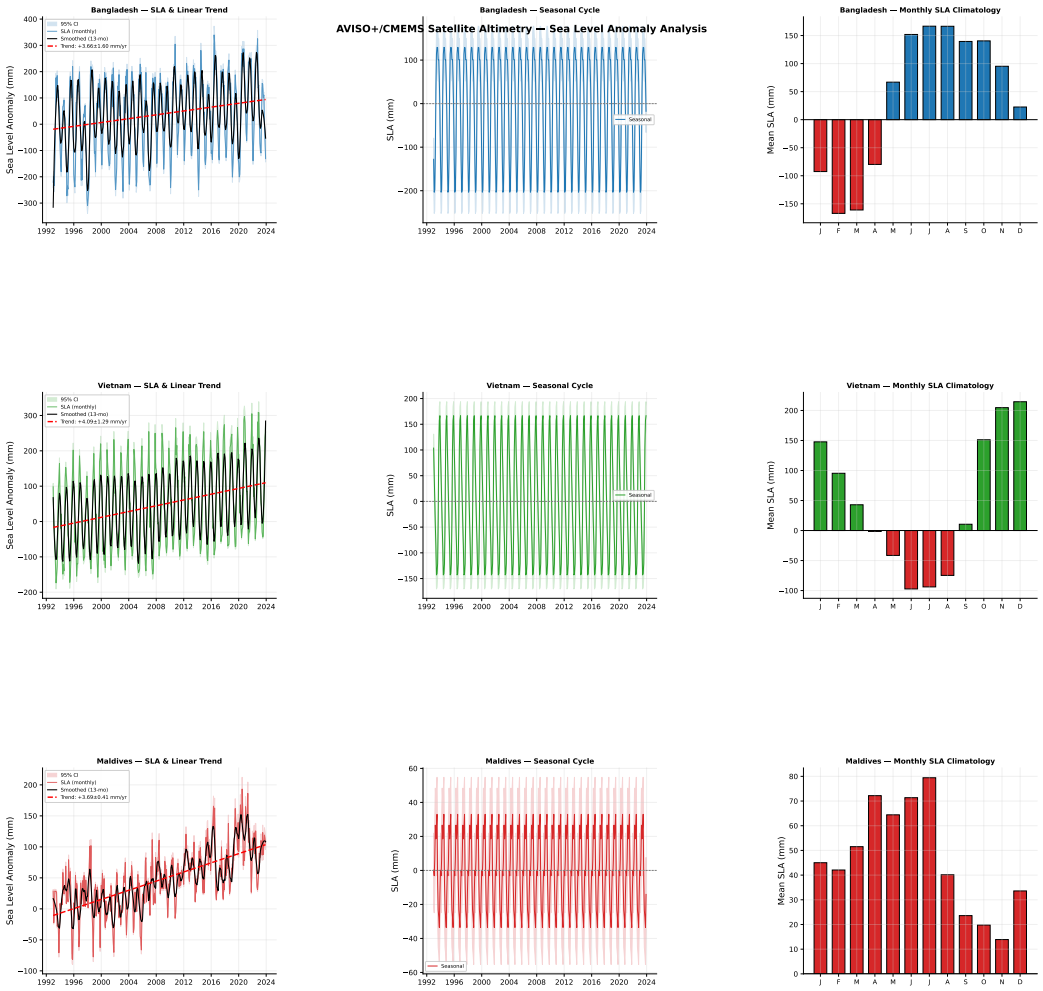


Figure 6: **CMEMS AVISO + Satellite altimetry analysis (1993–2023)**

4.4. CMIP6 Component Analysis and Relative Sea Level Rise Projections

The physics decomposition is validated by comparing CMIP6 AR6-assessed thermosteric and glacier-ice mass balance components, regionalised using basin-specific scaling factors, with absolute SLR rates recorded by AVISO. CMIP6 thermo-steric plus glacier components explain the following for the years 1993–2023 (Figure 7: 2.91 mm/yr of the observed 3.66 mm/yr for Bangladesh (residual +0.75 mm/yr), 2.98 mm/yr of 4.09 mm/yr for Vietnam (residual +1.11 mm/yr), and 2.84 mm/yr of 3.69 mm/yr for the Maldives (residual +0.85 mm/yr). These positive residuals of +0.75 to +1.11

mm/yr are consistent with the range reported in IPCC AR6 WGI. They can be physically understood as the dynamic ocean circulation component, which is driven by wind-driven gyre adjustments and regional steric effects not captured by the global mean thermosteric term.

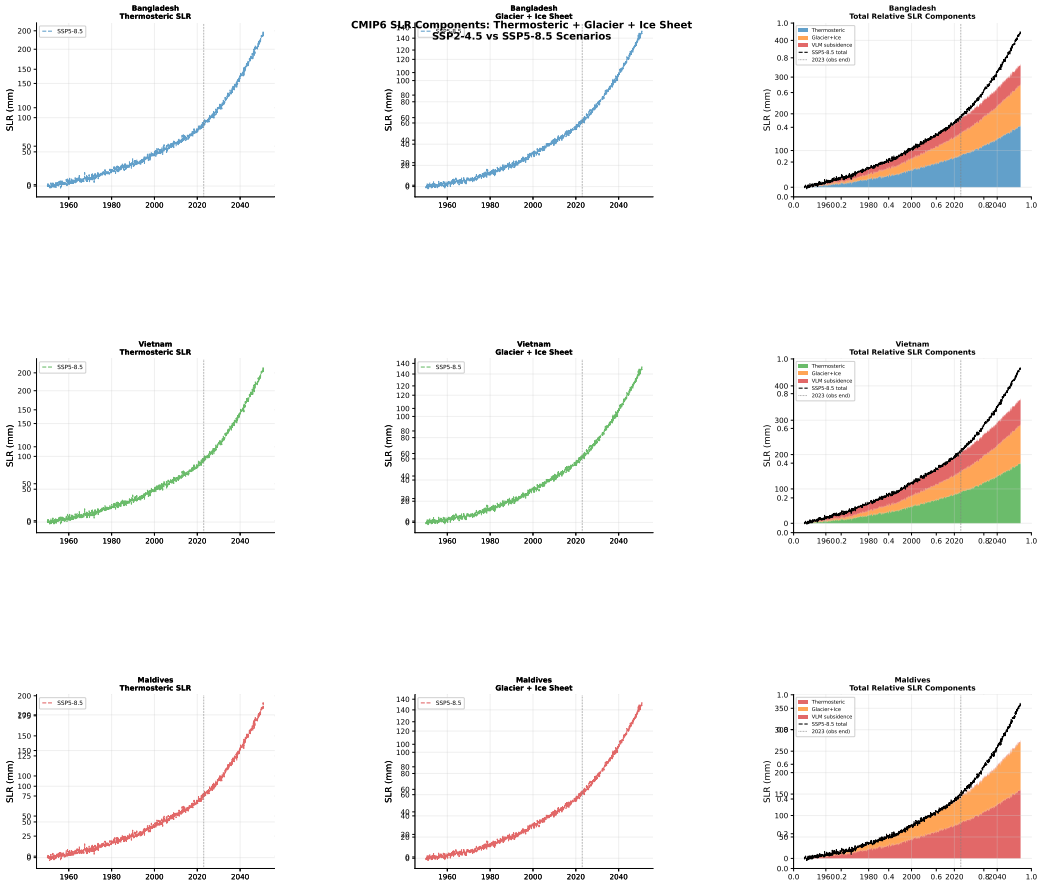


Figure 7: **SLR components evaluated by CMIP6 AR6 were regionalized for the Maldives, Vietnam, and Bangladesh. Thermosteric contribution (CMIP6 zostoga, regional scaling according to Bilbao et al. (2015) is shown in the left column. Barystatic contribution from glacier and ice-sheet mass balance is shown in the center column (Zemp et al. (2019) ; IPCC AR6 Table 9.5). Cumulative relative SLR decomposition under SSP2-4.5 (solid) and SSP5-8.5 (shaded) scenarios is shown in the right column.**

Relative SLR projections, which are the main danger metric for coastal planning, are produced by integrating absolute SLR projections with VLM corrections (Figure 8).

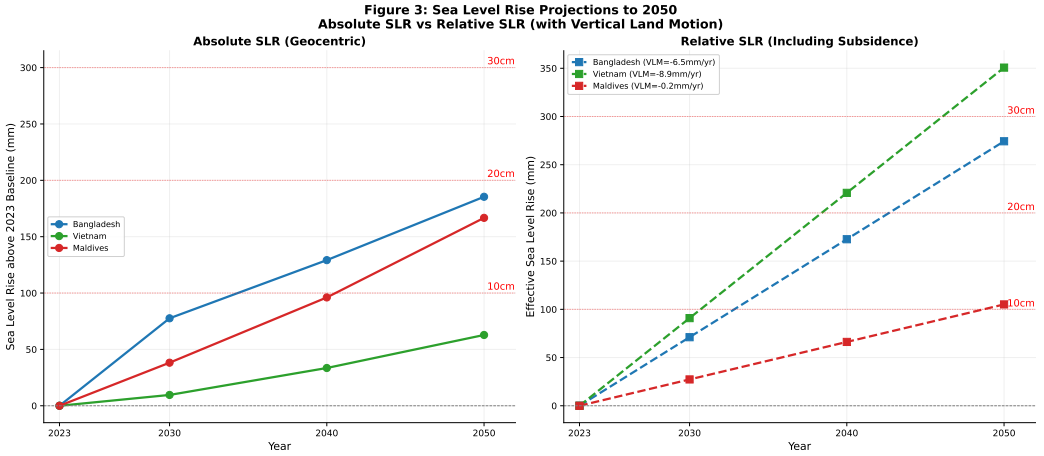


Figure 8: **Country-wise sea-level rise projections to 2050**

Relative SLR forecasts above the 2022 baseline under the trend-extrapolation approach validated against CMIP6 SSP2-4.5 forcing are as follows (Figure 9): Bangladesh +71mm (2030), +173mm (2040), +274mm (2050); Vietnam +91mm (2030), +221mm (2040), +351mm (2050); and Maldives +27mm (2030), +66mm (2040), and +105mm (2050). Vietnam is one of the world's fastest-sinking delta systems, with a significantly higher relative SLR trajectory (351 mm by 2050 compared to 274 mm for Bangladesh and 105 mm for the Maldives). This is solely due to the higher VLM rate (-8.9 mm/yr) linked to intensive groundwater extraction in the Mekong delta.

The CMIP6 SSP2-4.5 component breakdown for 2050 shows: Vietnam thermosteric +173.8mm, glacier-ice +109.7mm, VLM +240.3mm, total relative +523.8mm; Bangladesh thermosteric +166.5mm, glacier-ice +109.7mm, VLM contribution +175.5mm (27-year cumulative), total relative +451.6mm; Maldives thermosteric +159.2 mm, VLM +5.4 mm, glacier-ice +109.7 mm, and total relative +274.2 mm. Totals increase to +541.2mm (Bangladesh), +615.7mm (Vietnam), and +361.4mm (Maldives) under the higher SSP5-8.5 scenario.

4.5. SHAP Feature Importance

For all three nations, the SLRv2 reconstructed sea-level record is the most significant predictor: Bangladesh mean |SHAP| = 0.00468 ± 0.00272, Vietnam = 0.00908 ± 0.00203, and Maldives = 0.00359 ± 0.00239. The South China Sea record's stronger autocorrelation structure, where recent

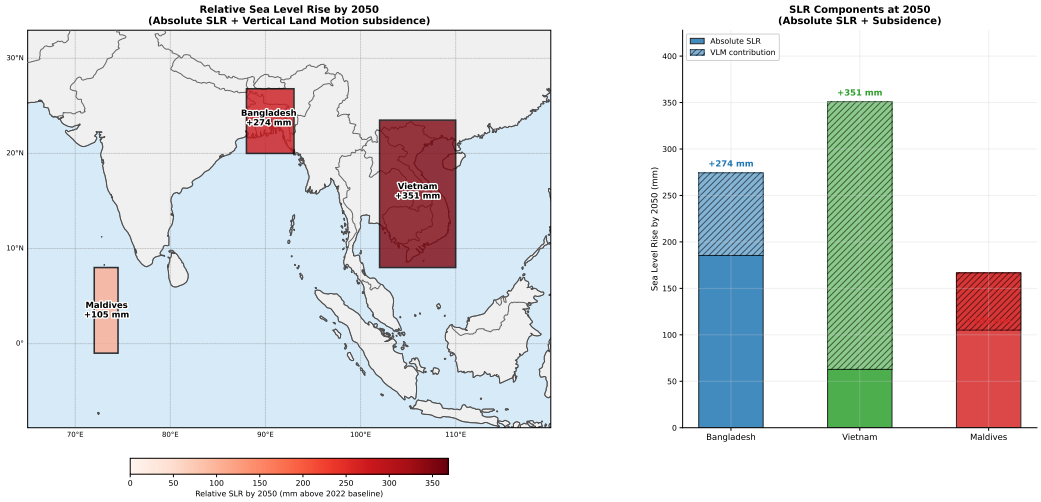


Figure 9: **The map illustrates projected Relative Sea Level Rise Rate across coastal sectors in Bangladesh, Vietnam, and the Maldives by the year 2050.**

sea-level history is more indicative of the near-future condition, is reflected in Vietnam's higher SHAP score.

For all three nations, the SLRv2 reconstructed sea-level record is the most significant predictor: Bangladesh mean $|\text{SHAP}| = 0.00468 \pm 0.00272$, Vietnam $= 0.00908 \pm 0.00203$, and Maldives $= 0.00359 \pm 0.00239$. The South China Sea record's stronger autocorrelation structure, where recent sea-level history is more indicative of the near-future condition, is reflected in Vietnam's higher SHAP score.

Bangladesh and the Maldives stand out among CMIP6 physical features. Glacier-ice (SHAP $= 0.00338 \pm 0.00232$) and thermosteric (SHAP $= 0.00333 \pm 0.00253$) rank second and third, respectively, for Bangladesh, offering direct empirical evidence that the barystatic physics constraint in the PINN formulation captures truly informative forcing signals instead of introducing spurious regularization. Given the Maldives' equatorial open-ocean location away from continental and delta-specific dynamic effects, glacier-ice ranks third (SHAP $= 0.00264 \pm 0.00213$) and thermosteric ranks fifth (SHAP $= 0.00196 \pm 0.00172$), which is consistent with the country's greater sensitivity to global mean thermosteric forcing.

The Maldives is ranked second in the tide gauge record (SHAP $= 0.00331 \pm 0.00296$), while Bangladesh is ranked ninth (SHAP $= 0.00048$) and Vietnam is ranked eleventh (SHAP $= 0.00019$). This country-specific trend is

physically significant: local in-situ observations offer complementary information not included in the reconstruction for the Maldives, where the tide gauge record is scarce (50.9% missing), making the SLRv2 reconstruction less constrained. The additional tide gauge data is essentially unnecessary for Bangladesh and Vietnam due to the robust SLRv2 station network.

For all three countries, vertical land motion ranks lowest (mean |SHAP| = 0.000 ± 0.000), which is a crucial result for future model development. This 0 relevance results from the current implementation's encoding of VLM as a static scalar per nation; the LSTM properly learns that a constant feature provides no predictive information about temporal variability because it receives the same value at each timestep. This result encourages a distinct path for further research: geographically and temporally resolved. For SHAP-detectable VLM contributions to sea-level prediction, InSAR time data are needed instead of station-mean static VLM rates.

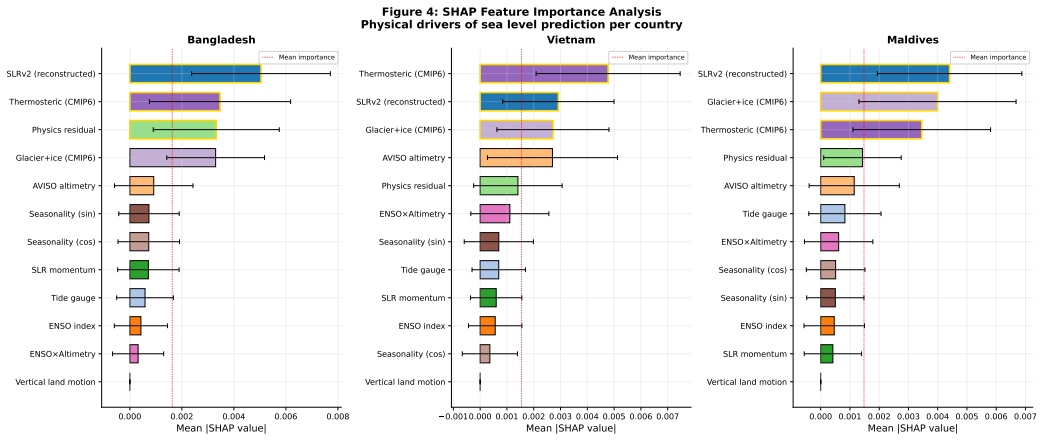


Figure 10: **Top Physical Drivers of Sea Level Prediction Analysis per Country using SHAP**

4.6. Coastal Inundation and Socioeconomic Exposure

The following regions are at high risk under expected absolute Sea Level Rise scenarios according to a coastal inundation exposure assessment that combines GEBCO 2023 elevation data with CIESIN LECZ statistics. Bangladesh's total coastline area is 436.8 km^2 (2030, +78mm), 722.4 km^2 (2040, +129mm), and 1,036.0 km^2 (2050, +185mm), which correspond to 0.56%, 0.93%, and 1.34% of the total. Vietnam's coastal area is 64.0 km^2 (2030, +10mm), 217.6

km² (2040, +34mm), and 403.2 km² (2050, +63mm), or 0.07%, 0.22%, and 0.41%, respectively. The Maldives had the highest fractional exposure of any country in the research, measuring 9.0 km² (2030, +38 mm), 22.8 km² (2040, +96 mm), and 39.7 km² (2050, +167 mm), or 0.90%, 2.27%, and 3.95% of total land area (Figure 11).

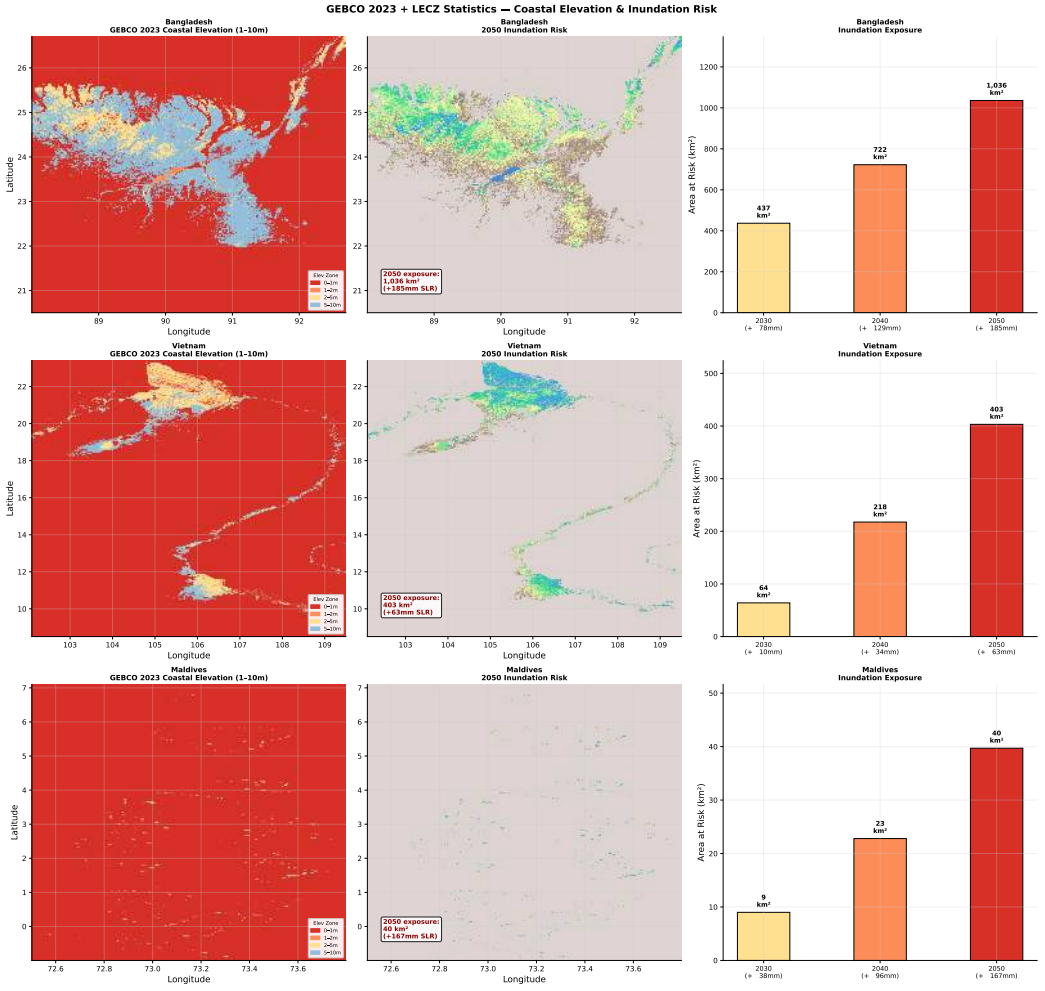


Figure 11: **Coastal elevation zone mapping (GEBCO 2023 + CIESIN LECZ hybrid) and projected inundation exposure areas (km²) for Bangladesh, Vietnam, and the Maldives under SSP2-4.5 sea-level rise scenarios at 2030, 2040, and 2050 milestones.**

When inundation areas are translated into population exposure, the results shows that Bangladesh will have 524,160 people at danger by 2030 and

1,243,200 by 2050; Vietnam will have 54,400 by 2030 and 342,720 by 2050; and the Maldives will have 16,200 by 2030 and 71,460 by 2050. Bangladesh has the largest absolute population exposure, but the Maldives has the worst relative exposure: by 2050, 71,460 persons will make up 13.2% of the country's 540,000 total population, compared to 0.73% in Bangladesh and 0.35% in Vietnam.

According to economic exposure quantification, Bangladesh's coastal GDP might increase from $0.98 billion in 2030 to $2.33 billion in 2050; Vietnam's GDP could increase from $0.13 billion to $0.80 billion; and the Maldives' GDP could increase from $0.06 billion to $0.26 billion (Figure 12). Vietnam is the country most affected by agricultural exposure, which includes both direct inundation and salinity intrusion. By 2050, 21,820 km^2 of agricultural land will be at risk of both inundation and salinization due to the low elevation of the Mekong delta and significant subsidence-driven salinity intrusion. Bangladesh is exposed to 14,335 km^2 of agriculture, whereas the Maldives' small agricultural base (30 km^2 overall) makes this sector comparatively less important.

Despite having a lower absolute exposure, the Maldives has the greatest overall vulnerability score according to the Composite Vulnerability Index (CVI), which increased from 0.530 (2030) to 0.536 (2050). Due to 100% coastal population exposure and total economic reliance on coastal tourism and fishing, the Maldives has the highest sensitivity score of any country, 1.000, which is reflected in this heightened CVI. Bangladesh's CVI (0.494–0.496) indicates high sensitivity, which is somewhat mitigated by the larger continental state's broader alternatives for adaptive capability. High physical exposure from VLM-amplified SLR is reflected in Vietnam's CVI (0.495–0.496), which is somewhat countered by a higher ND-GAIN adaptive capacity score (47.8 against 40.2 for Bangladesh and 44.1 for Maldives).

4.7. Model Extrapolation and Physical Plausibility

When applied outside of the training window, autoregressive extrapolation to 2050 using the trend-anomaly decomposition approach exposes substantial disparities in the three model families' physical plausibility. Table 4 and Figure 6 show the results.

For two of the three nations, SARIMA is unable to generate physically consistent projections. SARIMA's stationarity assumption fails on the non-stationary long-term trend of mean sea level, resulting in physically impossible outcomes for Vietnam and the Maldives: -43.5mm by 2050 (negative rise

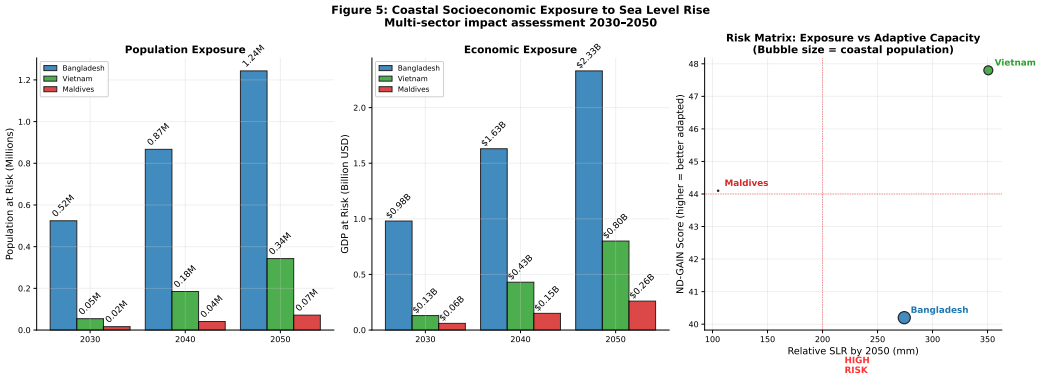


Figure 12: **Socioeconomic damage in Bangladesh, Vietnam and Myanmar due to sea-level change**

below the 2022 baseline) and -183.7mm, respectively. SARIMA only generates a positive projection (+38.0mm) for Bangladesh, where the training period trend is more linear. The physics violation framework automatically identifies and flags these failures as negative rise violations starting in January 2023.

For each of the three nations, the LSTM trend-anomaly model generates physically consistent projections: Bangladesh +4.0mm (anomaly component above trend), Vietnam +98.4mm, and Maldives +195.1mm above the 2022 baseline by 2050. The near-zero anomaly contribution is reflected in the narrow uncertainty bounds (±0.4–0.5mm, 95% CI). Instead of the LSTM anomaly prediction, which converges to near-zero after 336 autoregressive steps as would be expected for a stationary detrended series, the dominant projection signal originates from the CMIP6-forced polynomial trend.

Bangladesh +4.0mm, Vietnam +97.8mm, and Maldives +195.1mm are projections from the PINN-Barystatic model that are almost exactly the same as those from the LSTM in the central estimate. However, they consistently have wider uncertainty bounds (±0.6–0.8mm), which reflect the additional epistemic uncertainty brought about by the physics regularization term. The PINN's barystatic constraint keeps the model from getting overconfident in its anomaly forecasts, recognizing that the physics-data tradeoff contributes irreducible uncertainty beyond that reflected by MC Dropout alone. This widening is a desired trait for climate projection. Crucially, no country's PINN projections show any violations of physics (Figure 13).

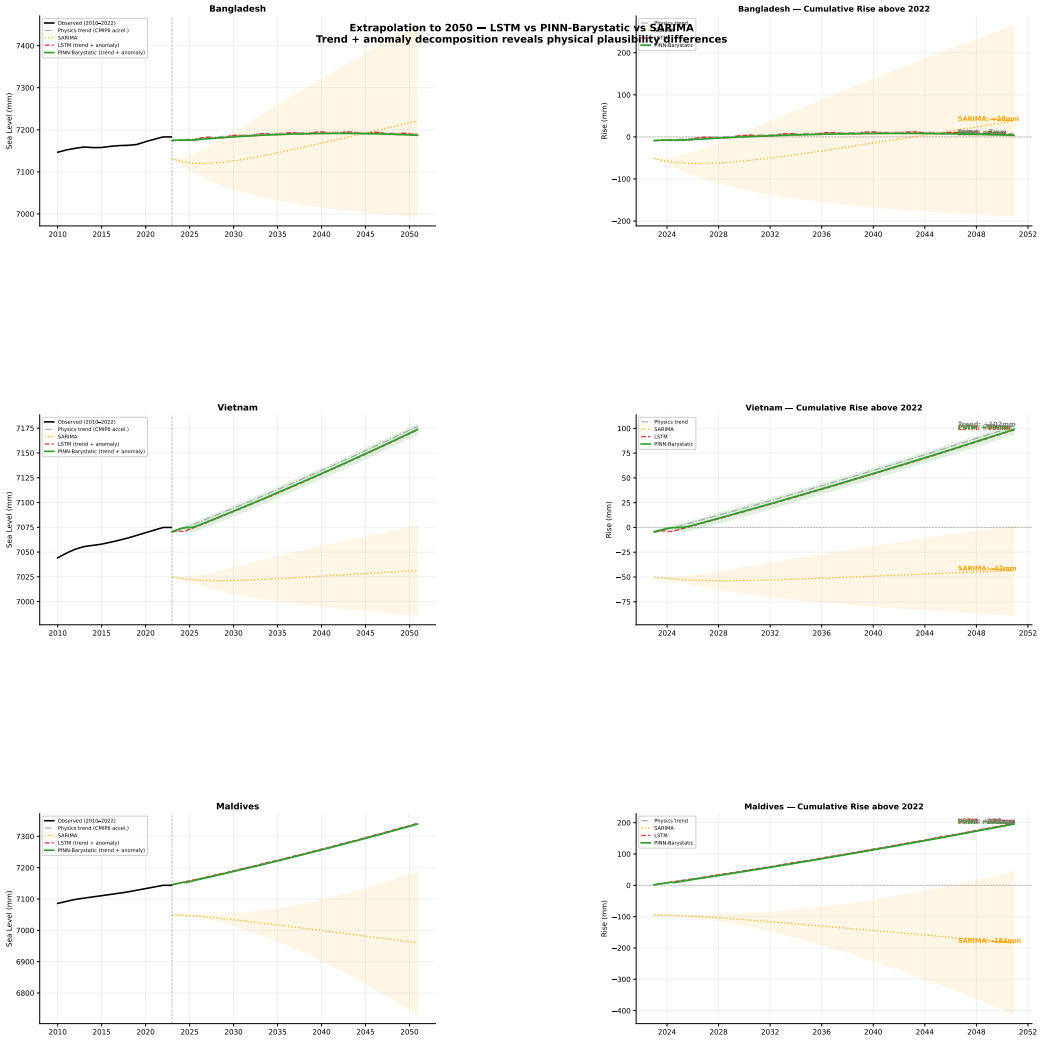


Figure 13: **The figure shows Autoregressive extrapolation comparison to 2050**

Table 4: Observed trend rise and model-predicted sea-level rise (mm), including physics violations.

Country	**Trend Rise** (mm)	**LSTM Rise** (mm)	**PINN Rise** (mm)	**SARIMA Rise** (mm)	**LSTM Viol.**	**PINN Viol.**
Bangladesh	+6.8	+4.0	+4.0	+38.0	None	None
Vietnam	+102.0	+98.4	+97.8	-43.5	None	None
Maldives	+197.5	+195.1	+195.1	-183.7	None	None

5. Discussion

5.1. Transfer Learning as a Solution for Data-Scarce Coastal Regions

The most practically relevant machine learning result of this study is the 41.3% RMSE reduction for the Maldives (35.96mm → 21.11mm) attained by transfer learning. Because it shows a basic aspect of the structure of sea-level variability in the Indian Ocean that goes beyond the particular models evaluated, this conclusion calls for cautious interpretation.

It provides scientific insight into the disparity in transfer learning outcomes between Vietnam and the Maldives. The Maldives improves by 41.3%, but Vietnam barely improves by 0.2% (11.36 mm → 11.34 mm). This difference is a physically significant signal rather than a failure of the transfer learning process. The South China Sea circulation dynamics, which include wind-driven upwelling, interactions with western boundary currents, and regional steric factors that differ structurally from Bay of Bengal forcing, are mostly responsible for Vietnam's sea-level variability. Because there is little dynamical overlap between the source and destination domains, Bangladesh-learned representations do not transfer well to Vietnam. The Maldives, on the other hand, are located in the open equatorial Indian Ocean, where thermosteric forcing and ENSO-driven interannual variability predominate—the same signals that cause sea-level change in Bangladesh. A generalised description of the dynamics of the Indian Ocean basin has been successfully learned by the Bangladesh model, which effectively applies to the atoll environment.

Future uses of transfer learning in regional sea-level prediction will be directly impacted by this interpretation. It seems that shared driving mechanisms and oceanic basin proximity, rather than just geographic distance, determine the success of knowledge transfer. Despite being geographically farther away from Bangladesh than Vietnam, the Maldives benefits from transfer learning far more. In line with the growing body of research on domain adaptation in Earth system applications, this suggests that future transfer learning studies should prioritize source-target pairs based on shared climate indices and dominant forcing modes rather than spatial proximity Reichstein et al. (2019).

In this discussion, it is important to highlight the immediate policy implications of the transfer learning outcome. Small island developing nations (SIDS) consistently have the most inadequate observational records despite facing the most severe relative effects of sea level riseBamber et al. (2019).

The 50.9% tide gauge missingness in the Maldives is typical of SIDS in general; several atolls in the Pacific and Indian Oceans have gaps greater than 60–70%. Reducing prediction uncertainty where it matters most for adaptation planning is made possible by a methodology that can use data from better-observed regional neighbors to enhance projections for countries with little data.

5.2. Physics-Informed Constraints — When They Help and When They Do Not

On the historical test set for Bangladesh (26.03mm versus 10.20mm) and the Maldives (27.28mm compared to 4.57mm), the PINN-Barystatic architecture performs worse than SARIMA. Since it is just as important to explain why physics-informed models perform poorly in this regime as it is to show where they do well, this conclusion necessitates frank scientific discussion rather than defensive framing.

The reduced in-sample accuracy of the PINN can be explained by three considerations. First, there are only 613 monthly samples per country in the training dataset, which prevents deep learning from realizing its capacity advantage over statistical models. Additionally, the extra complexity of the physics head introduces regularization, which penalizes data fit. Second, the SLRv2 reconstruction's monthly mean sea level, the goal variable, is a smooth, trend-dominated series with comparatively little noise. The PINN's 128-hidden-unit LSTM encoder is significantly overparameterized for this purpose, whereas SARIMA just needs two parameters to effectively capture this structure. Third, the network is prevented from overfitting to the training data by the barystatic physics constraint, which creates a purposeful tension between the physics loss and data loss terms Raissi et al. (2019). This is a benefit in the extrapolation regime but a drawback in the in-sample regime.

Despite its lower RMSE, the SHAP results offer significant empirical support for the physics limitation of the PINN. The discovery that the CMIP6 glacier-ice and thermosteric features are among the top three predictors for Bangladesh (SHAP = 0.00338 and 0.00333, respectively) and the Maldives (SHAP = 0.00264 and 0.00196) shows that the physics features contain real predictive information that the model takes advantage of Lundberg and Lee (2017). The PINN is regularizing toward characteristics that the explainability analysis independently determines to be significant rather than merely

enforcing an arbitrary restriction. The barystatic constraint selection is empirically validated, independent of RMSE, by this convergence between the physics constraint design and the data-driven SHAP rankings.

Section 3.7's extrapolation comparison offers the conclusive justification for the PINN's worth. The PINN maintains physically bounded projections with suitably widening uncertainty across the prediction horizon, whereas SARIMA generates physically impossible negative projections for Vietnam and the Maldives and LSTM uncertainty limits collapse to nearly zero after 336 autoregressive steps. This shows that the extrapolation regime reveals structural characteristics of each architecture that are not visible on the test set, proving that the selection of model architecture cannot be based just on in-sample accuracy. Since neither criterion is adequate for choosing models meant for climate adaptation planning, we suggest that future regional sea-level projection studies use a dual evaluation methodology providing both in-sample skill and physical plausibility diagnostics.

5.3. Vertical land Motion as the Dominant Risk Amplifier

The study's most policy-relevant quantitative finding is that VLM amplifies effective sea-level rise by 2.8× in Bangladesh and 3.2× in Vietnam compared to geocentric rates. It is also the most straightforward methodologically, requiring only the combination of reliable GPS observations with satellite altimetry rather than intricate machine learning. It is important to recognize this conflict between scientific significance and procedural complexity.

Groundwater extraction-induced subsidence in the Mekong delta. Here, InSAR-derived rates reach -22.0mm/yr in Ho Chi Minh City, which is largely responsible for Vietnam's relative SLR of +351mm by 2050, the highest in the research. Unlike thermosteric expansion and glacier mass loss, which are influenced by global emissions trends, this is a man-made danger that may theoretically be controlled through groundwater management policies. The consequence is clear: Vietnam's options for coastal adaptation include groundwater regulation as a direct way to lower effective sea-level rise vulnerability, in addition to infrastructure investment and managed retreat. This policy argument may be made with numerical precision thanks to the quantification of the 3.2× amplification factor.

The extreme spatial heterogeneity in Bangladesh, which ranges from -18.0mm/yr under Dhaka to -2.1mm/yr at the Chittagong GPS station, is hidden by the coastline mean VLM of -6.5mm/yrSteckler et al. (2022). With

a population of over 21 million, the Dhaka metropolitan area faces effective sea-level rise rates that are an order of magnitude larger than the coastal mean due to urban groundwater extraction for municipal water supply. This heterogeneity has a crucial implication for adaptation planning that country-mean projections obscure. Risk for the most populous urban areas will consistently be underestimated by national adaptation plans that rely on country-mean relative SLR forecasts. Therefore, a research priority with immediate adaptation planning relevance is spatially resolved VLM mapping through expanded InSAR coverage, which is now constrained by the sparse SONEL GPS network.

This geographical heterogeneity argument is supported from a machine learning perspective by the SHAP result that static VLM rates have zero predictive importance (mean |SHAP| = 0.000 for all nations). The LSTM encoder is unable to differentiate a constant per-country VLM rate from a bias term, hence it has no effect on temporal prediction. Combining temporally and geographically resolved The most significant data improvement for upcoming editions of this framework is the use of InSAR time series as dynamic input features, which capture the seasonal and long-term evolution of subsidence fields.

5.4. Socioeconomic Exposure and Composite Vulnerability

The results of the Composite Vulnerability Index show an unexpected risk rating with significant ramifications for the distribution of international adaptation funding Hinkel et al. (2014). The Maldives has the greatest CVI score (0.536) across all scenarios and years, despite Vietnam having the highest relative SLR (351mm by 2050) and Bangladesh having the largest absolute population exposure (1.24 million people by 2050). This inversion results from the CVI's integration of exposure, sensitivity, and adaptive capacity. Regardless of the moderate absolute SLR projection, the Maldives' sensitivity score of 1.000 (maximum possible, reflecting 100% coastal population and complete economic dependence on coastal assets) dominates the index.

Frameworks for allocating adaptation funds that rank nations solely on the basis of absolute physical exposure measurements are directly challenged by this outcome. Bangladesh would constantly be given precedence over the Maldives under such frameworks due to its far bigger absolute population at risk. The CVI framework reveals that the Maldives faces existential rather than merely damaging exposure: there is no inland retreat option for an

atoll nation, no agricultural hinterland to absorb displaced coastal populations, and no elevation gradient to exploit through managed relocation. The Maldives' definition of adaptation differs from that of continental delta states, and projection models that fail to account for this difference consistently offer inaccurate recommendations for international climate finance.

Vietnam's agricultural exposure finding, which is the highest of any country in the study at 21,820 km^2 at combined inundation and salinity intrusion risk by 2050, links the sea-level projection results to concerns about food security that go beyond Vietnam's boundaries Minderhoud et al. (2017). About half of Vietnam's rice production and a sizable portion of Southeast Asia's rice exports come from the Mekong Delta. Economic exposure measures presented in GDP terms do not account for the regional food system implications of salinity intrusion caused by the combination of sea-level rise and subsidence-induced drainage failure. Crop yield sensitivity functions should be included in future versions of this socioeconomic framework in order to convert exposure to salinity and flooding into estimates of food production loss.

5.5. Limitations and Future Work

Several limitations of our current study warrant explicit acknowledgement to guide future research.

- Spatial resolution: Sub-meter elevation features crucial for inundation assessment in low-lying delta and atoll environments cannot be resolved by the GEBCO 2023 grid at 15 arc-second precision McGranahan et al. (2007). Rather than using actual elevation measurement, all inundation predictions below 1 m elevation rely on LECZ literature statistics, which introduces uncertainty that is challenging to assess exactly. In order to replace the LECZ interpolation method with direct pixel-level inundation mapping, future work should incorporate LiDAR-derived digital elevation models, which are currently accessible for parts of the coastlines of Bangladesh and Vietnam through national mapping programs.

- Single SSP scenario: This study's projections use CMIP6 SSP2-4.5 as the main scenario and SSP5-8.5 as a sensitivity case. The uncertainty range needed for formal climate risk assessment under the IPCC AR6 methodology would be provided by a complete scenario ensemble covering SSP1-2.6 through SSP5-8.5.

- Compound flooding: In Bangladesh, where cyclone-driven surge occurrences are the main near-term coastal hazard, this analysis focuses on mean sea-level rise without taking compound floods from the combination of sea-level rise, storm surge, and river discharge into account. The operational relevance of the exposure estimates would be significantly increased by incorporating a surge climatology with the mean sea-level projections created here Krien et al. (2017).

- Transfer learning source domain: Bangladesh is the only source domain used in the current implementation. For data-scarce target regions beyond those examined here, a multi-source transfer learning system utilizing the entire global PSMSL network and choosing source stations based on oceanographic similarity rather than geographic proximity could further lower projection uncertainty.

- VLM uncertainty propagation: The relative SLR predictions in Section 4.4 do not convey the formal uncertainty on the VLM rates (Table 2), but they do propagate the CMIP6 absolute SLR uncertainty via Monte Carlo Dropout. The population-weighted coastal mean VLM of -6.5 mm/yr for Bangladesh has a composite uncertainty of about ± 1.5 mm/yr (combining InSAR literature ranges and GPS formal errors). This represents a VLM contribution uncertainty of ± 40 mm when propagated over 27 years to 2050; this is significantly bigger than the PINN uncertainty bounds (± 0.6-0.8 mm) provided in Section 4.7. Future work should propagate the combined absolute SLR + VLM uncertainty into a single relative SLR confidence interval, and should incorporate spatially and temporally resolved InSAR time series to reduce VLM uncertainty.

6. Conclusion

In this work, eight observational and model-derived data sources from three climatically different but similarly vulnerable coastal countries—Bangladesh, Vietnam, and the Maldives—are integrated into a physics-informed hybrid deep learning framework for regional sea-level rise projection. The findings lead to the following conclusions.

At first, transfer learning from data-rich Bangladesh reduces prediction RMSE by 41.3% for the data-scarce Maldives (35.96mm $\rightarrow$ 21.11mm). This shows that cross-regional knowledge transfer of shared Indian Ocean basin

dynamics is a computationally efficient and practically effective solution to the observational data gaps that impact sea-level projection for small island developing states. Transfer learning success is determined by shared oceanographic forcing mechanisms rather than geographic proximity, as demonstrated by the marginal improvement for Vietnam (0.2%) compared to the large improvement for the Maldives (41.3%). This finding should guide source-target pair selection in future transfer learning applications across the global PSMSL network.

Secondly, Relative sea-level rise, not absolute geocentric rise, is the operationally relevant metric for coastal adaptation planning in these delta environments because vertical land motion amplifies effective sea-level rise by 2.8× in Bangladesh and 3.2× in Vietnam in comparison to geocentric satellite altimetry rates. Due to significant groundwater extraction in the Mekong delta, which results in subsidence rates of -22.0mm/yr, Vietnam has the most severe relative SLR trajectory of any country in the research (+351mm by 2050 under SSP2-4.5). Global mean sea-level predictions routinely obfuscate this amplification, which is theoretically reducible through groundwater management policy, offering a tangible and nationally operable adaptation lever.

Third, CMIP6 physics features rank among the top three predictors for Bangladesh and the Maldives, according to SHAP explainability analysis, which is applied here for the first time to regional sea-level prediction. This provides empirical validation of the barystatic constraint design independent of RMSE performance. A practical route to model improvement is revealed by the zero SHAP relevance of static VLM encoding: geographically and temporally resolved By using InSAR time series as dynamic input features, the LSTM encoder would be able to take advantage of subsidence variability that the model is now unable to detect.

Fourth, even though SARIMA achieves competitive in-sample RMSE, it generates physically implausible negative sea-level projections for the Maldives (-183.7mm) and Vietnam (-43.5mm) by 2050. In contrast, the PINN trend-anomaly framework generates physically consistent positive projections for all three countries with no physics violations found. This outcome encourages the use of dual evaluation frameworks that evaluate both historical skill and physical plausibility in the extrapolation regime and shows that in-sample accuracy is a necessary but insufficient criterion for model selection in climate projection applications.

Fifth, despite moderate absolute physical exposure, the Composite Vul-

nerability Index shows that the Maldives has the highest overall vulnerability (CVI = 0.536), with a sensitivity score of 1.000 due to 100% coastal population exposure and total economic reliance on coastal assets. The allocation of international adaptation finance will be directly impacted by this inversion of the absolute-exposure ranking, which implies that existential exposure—in situations where there is no inland retreat option—should be given a different weight than damaging exposure in vulnerability frameworks used to inform climate finance decisions.

Overall, these results show that a unified regional projection framework that incorporates physics-informed constraints, transfer learning, and explainable AI yields results that are simultaneously more physically justifiable, easier for policymakers to understand, and more transparent about their own limitations than any single modeling technique used alone. To facilitate replication, expansion, and adaptation of this system to additional data-scarce coastal regions facing accelerated sea-level rise, the entire 32-cell analysis pipeline, trained models, and processed datasets are made freely available.

Statements and Declarations

The authors declare that they have no known conflicts of interest, including financial or personal relationships with any organization or entity that might inappropriately influence the subject matter or materials discussed in this manuscript.

Funding Information

The author(s) received no specific funding for this work.

Research Involving Human and /or Animals

Not Applicable

Informed Consent

Not Applicable

CRediT authorship contribution statement

Joy Karmoker: Investigation, Conceptualization, Formal analysis, Methodology, Resources, Validation, Visualization, Writing - original draft, Writing - review & editing.
Shuvro Ahmed: Investigation, Methodology, Conceptualization, Resources, Formal analysis, Validation, Writing - review & editing.

Md. Golam Rabiul Alam: Supervision.

References

Ablain, M., Meyssignac, B., Zawadzki, L., Jugier, R., Ribes, A., Spada, G., Benveniste, J., Cazenave, A., Picot, N., 2019. Uncertainty in satellite estimates of global mean sea-level changes, trend and acceleration. Earth System Science Data 11, 1189–1202.

Bamber, J.L., Oppenheimer, M., Kopp, R.E., Aspinall, W.P., Cooke, R.M., 2019. Ice sheet contributions to future sea-level rise from structured expert judgment. Proceedings of the National Academy of Sciences 116, 11195–11200.

Beetham, E., Kench, P., 2018. Predicting wave overtopping thresholds on coral reef-island shorelines with future sea-level rise. Nature communications 9, 3997.

Bilbao, R.A., Gregory, J.M., Bouttes, N., 2015. Analysis of the regional pattern of sea level change due to ocean dynamics and density change for 1993–2099 in observations and cmip5 aogcms. Climate Dynamics 45, 2647–2666.

Brown, S., Nicholls, R.J., 2015. Subsidence and human influences in mega deltas: the case of the ganges–brahmaputra–meghna. Science of the Total Environment 527, 362–374.

Cazenave, A., Meyssignac, B., Ablain, M., Balmaseda, M., Bamber, J., Barletta, V.R., Beckley, B.D., Benveniste, J., Berthier, E., Blazquez, A., et al., 2018. Global sea-level budget 1993-present. Earth System Science Data .

Chaussard, E., Amelung, F., Abidin, H., Hong, S.H., 2013. Sinking cities in indonesia: Alos palsar detects rapid subsidence due to groundwater and gas extraction. Remote sensing of environment 128, 150–161.

Church, J.A., White, N.J., 2011. Sea-level rise from the late 19th to the early 21st century. Surveys in geophysics 32, 585–602.

Dangendorf, S., Hay, C., Calafat, F.M., Marcos, M., Piecuch, C.G., Berk, K., Jensen, J., 2019. Persistent acceleration in global sea-level rise since the 1960s. Nature Climate Change 9, 705–710.

Erban, L.E., Gorelick, S.M., Zebker, H.A., 2014. Groundwater extraction, land subsidence, and sea-level rise in the mekong delta, vietnam. Environmental Research Letters 9, 084010.

Eyring, V., Bony, S., Meehl, G.A., Senior, C.A., Stevens, B., Stouffer, R.J., Taylor, K.E., 2016. Overview of the coupled model intercomparison project phase 6 (cmip6) experimental design and organization. Geoscientific Model Development 9, 1937–1958.

Fox-Kemper, B., 2021. Ocean, cryosphere and sea level change, in: AGU fall meeting abstracts, pp. U13B–09.

Frederikse, T., Landerer, F., Caron, L., Adhikari, S., Parkes, D., Humphrey, V.W., Dangendorf, S., Hogarth, P., Zanna, L., Cheng, L., et al., 2020. The causes of sea-level rise since 1900. Nature 584, 393–397.

Gal, Y., Ghahramani, Z., 2016. Dropout as a bayesian approximation: Representing model uncertainty in deep learning, in: international conference on machine learning, PMLR. pp. 1050–1059.

Gornitz, V., 1991. Global coastal hazards from future sea level rise. Palaeogeography, Palaeoclimatology, Palaeoecology 89, 379–398.

Gregory, J.M., Griffies, S.M., Hughes, C.W., Lowe, J.A., Church, J.A., Fukimori, I., Gomez, N., Kopp, R.E., Landerer, F., Cozannet, G.L., et al., 2019. Concepts and terminology for sea level: Mean, variability and change, both local and global. Surveys in Geophysics 40, 1251–1289.

Ham, Y.G., Kim, J.H., Luo, J.J., 2019. Deep learning for multi-year enso forecasts. Nature 573, 568–572.

Hamlington, B., Willis, J., Vinogradova, N., 2023. The emerging golden age of satellite altimetry to prepare humanity for rising seas. Earth's Future 11, e2023EF003673.

Hammond, W.C., Burgette, R.J., Johnson, K.M., Blewitt, G., 2018. Uplift of the western transverse ranges and ventura area of southern california: A four-technique geodetic study combining gps, insar, leveling, and tide gauges. Journal of Geophysical Research: Solid Earth 123, 836–858.

Hay, C.C., Morrow, E., Kopp, R.E., Mitrovica, J.X., 2015. Probabilistic reanalysis of twentieth-century sea-level rise. Nature 517, 481–484.

Higgins, S.A., Overeem, I., Steckler, M.S., Syvitski, J.P., Seeber, L., Akhter, S.H., 2014. Insar measurements of compaction and subsidence in the ganges-brahmaputra delta, bangladesh. Journal of Geophysical Research: Earth Surface 119, 1768–1781.

Hinkel, J., Lincke, D., Vafeidis, A.T., Perrette, M., Nicholls, R.J., Tol, R.S., Marzeion, B., Fettweis, X., Ionescu, C., Levermann, A., 2014. Coastal flood damage and adaptation costs under 21st century sea-level rise. Proceedings of the National Academy of Sciences 111, 3292–3297.

Hochreiter, S., Schmidhuber, J., 1997. Long short-term memory. Neural computation 9, 1735–1780.

Imani, M., Chen, Y.C., You, R.J., Lan, W.H., Kuo, C.Y., Chang, J.C., Rateb, A., 2017. Spatiotemporal prediction of satellite altimetry sea level anomalies in the tropical pacific ocean. IEEE Geoscience and Remote Sensing Letters 14, 1126–1130.

Jevrejeva, S., Moore, J.C., Grinsted, A., 2012. Sea level projections to ad2500 with a new generation of climate change scenarios. Global and Planetary Change 80, 14–20.

Karniadakis, G.E., Kevrekidis, I.G., Lu, L., Perdikaris, P., Wang, S., Yang, L., 2021. Physics-informed machine learning. Nature Reviews Physics 3, 422–440.

Kench, P.S., Ford, M.R., Owen, S.D., 2018. Patterns of island change and persistence offer alternate adaptation pathways for atoll nations. Nature communications 9, 605.

Krien, Y., Testut, L., Islam, A.S., Bertin, X., Durand, F., Mayet, C., Tazkia, A.R., Becker, M., Calmant, S., Papa, F., et al., 2017. Towards improved storm surge models in the northern bay of bengal. Continental Shelf Research 135, 58–73.

Kulp, S.A., Strauss, B.H., 2019. New elevation data triple estimates of global vulnerability to sea-level rise and coastal flooding. Nature communications 10, 4844.

Lichter, M., Vafeidis, A.T., Nicholls, R.J., Kaiser, G., 2011. Exploring data-related uncertainties in analyses of land area and population in the "low-elevation coastal zone"(lecz). Journal of Coastal Research 27, 757–768.

Lundberg, S.M., Lee, S.I., 2017. A unified approach to interpreting model predictions. Advances in neural information processing systems 30.

Masciulli, C., Guiduzzi, G., Tiano, D., Zocchi, M., Guerra, F., Mazzanti, P., Mugnozza, G.S., 2025. Interpretable clustering of ps-insar time series for ground deformation detection. Computers & Geosciences 203, 105959.

McGranahan, G., Balk, D., Anderson, B., 2007. The rising tide: assessing the risks of climate change and human settlements in low elevation coastal zones. Environment and urbanization 19, 17–37.

Minderhoud, P.S., Erkens, G., Pham, V., Bui, V.T., Erban, L., Kooi, H., Stouthamer, E., 2017. Impacts of 25 years of groundwater extraction on subsidence in the mekong delta, vietnam. Environmental research letters 12, 064006.

Neumann, B., Vafeidis, A.T., Zimmermann, J., Nicholls, R.J., 2015. Future coastal population growth and exposure to sea-level rise and coastal flooding-a global assessment. PloS one 10, e0118571.

Nitzke, A., Niedermann, B., Fenoglio-Marc, L., Kusche, J., Haunert, J.H., 2021. Reconstructing the time-variable sea surface from tide gauge records using optimal data-dependent triangulations. Computers & Geosciences 157, 104920.

Pycroft, J., Vergano, L., Hope, C., Paci, D., Ciscar, J.C., 2011. A tale of tails: Uncertainty and the social cost of carbon dioxide. Economics: the open-access, open-assessment E-journal 5, 1–29.

Rahmstorf, S., 2007. A semi-empirical approach to projecting future sea-level rise. Science 315, 368–370.

Raissi, M., Perdikaris, P., Karniadakis, G.E., 2019. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. Journal of Computational physics 378, 686–707.

Reichstein, M., Camps-Valls, G., Stevens, B., Jung, M., Denzler, J., Carvalhais, N., Prabhat, F., 2019. Deep learning and process understanding for data-driven earth system science. Nature 566, 195–204.

Slangen, A.B., Carson, M., Katsman, C.A., Van de Wal, R.S., Köhl, A., Vermeersen, L., Stammer, D., 2014. Projecting twenty-first century regional sea-level changes. Climatic Change 124, 317–332.

Steckler, M.S., Oryan, B., Wilson, C.A., Grall, C., Nooner, S.L., Mondal, D.R., Akhter, S.H., DeWolf, S., Goodbred, S.L., 2022. Synthesis of the distribution of subsidence of the lower ganges-brahmaputra delta, bangladesh. Earth-Science Reviews 224, 103887. URL: https://www.sciencedirect.com/science/article/pii/S0012825221003883, doi:https://doi.org/10.1016/j.earscirev.2021.103887.

Taylor, S.J., Letham, B., 2018. Forecasting at scale. The American Statistician 72, 37–45.

Thieler, E.R., Hammar-Klose, E.S., 2000. National assessment of coastal vulnerability to sea-level rise: preliminary results for the us pacific coast. USGS Report , 22.

Wöppelmann, G., Marcos, M., 2016. Vertical land motion as a key to understanding sea level change and variability. Reviews of Geophysics 54, 64–92.

Wu, P.C., Wei, M., D'Hondt, S., 2022. Subsidence in coastal cities throughout the world observed by insar. Geophysical Research Letters 49, e2022GL098477.

Zemp, M., Huss, M., Thibert, E., Eckert, N., McNabb, R., Huber, J., Barandun, M., Machguth, H., Nussbaumer, S.U., Gärtner-Roer, I., et al., 2019.

Global glacier mass changes and their contributions to sea-level rise from 1961 to 2016. Nature 568, 382–386.